Transport Restructuring and Regional Development: Evidence from Argentina, 1960–1991

(Working title; alternatives under discussion)

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Abstract

Between 1960 and 1991, Argentina closed roughly 10,000 kilometers of railway lines while its paved road network nearly tripled (1954–1986), a restructuring driven by fiscal consolidation rather than regional policy. We estimate how district-level outcomes responded to the resulting changes in market access, computed from least-cost routes integrating both modes across 312 districts. To address endogenous network placement, we instrument with the technical closure rules of the 1962 Larkin Plan, a World Bank-backed assessment of the transport network, and with hypothetical road networks connecting major cities. Total population shows a small, marginally significant response (elasticity 0.052, SE 0.031); manufacturing responds strongly: 0.317 for production value, 0.378 for wage mass; agriculture shows no response. The population estimate, though not the sectoral contrast, is sensitive to the trade-elasticity calibration. Counterfactual decompositions attribute similar estimates to rail and road; roughly half of the response operates through own-district infrastructure rather than regional connectivity.

Keywords: Regional Development, Market Access, Transport Infrastructure, Railroads, Highways, Fiscal Consolidation, Argentina
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1 Introduction

How do large-scale transport infrastructure changes reshape regional economies? While most research examines infrastructure expansions designed to promote development, governments often face a different challenge: fiscal consolidation that forces difficult choices about which infrastructure to maintain. Argentina's experience between 1960 and 1991 provides a unique window into this question. Facing severe fiscal pressure, the government implemented a massive transport restructuring: roughly 10,000 kilometers of railway lines were closed, while the paved road network expanded from roughly 10,000 to 28,000 kilometers between 1954 and 1986. This episode offers rare evidence on how fiscal consolidation-driven infrastructure changes—rather than development-oriented investments—affect regional market access and economic outcomes.

This setting is valuable for two reasons. First, it speaks to a policy-relevant question: when governments must cut infrastructure spending, how do these cuts reshape the economic geography? The Argentine restructuring, motivated by fiscal rationalization rather than regional development goals, effectively acted as an unintended industrial and regional policy by shifting the dominant transport mode from rail to road. Second, it provides an opportunity to estimate how regional outcomes respond to market access changes in a multimodal infrastructure shock. Unlike typical infrastructure studies that examine single-mode expansions, Argentina's bundled rail-road restructuring allows us to study elasticities of population and sectoral economic activity to market access changes that integrate both transport modes.

Estimating the causal effects of such infrastructure changes faces two fundamental challenges. First, infrastructure placement and removal decisions are endogenous. Even when motivated by fiscal consolidation, governments may close lines in already-declining regions or expand roads where growth is expected. This makes it difficult to separate the causal effect of market access changes from pre-existing economic trends.

Second, when both rail and road networks change simultaneously, their effects are difficult to disentangle. The changes are often spatially correlated—regions losing rail access may or may not gain road access—creating multicollinearity problems if we try to estimate mode-specific effects in a single regression. Moreover, the theoretical framework suggests that what matters for regional outcomes is total market access integrating all transport modes, not the separate contribution of each mode. This raises a methodological question: what can and cannot be identified about mode-specific effects in multimodal infrastructure episodes?

This paper addresses these challenges by studying Argentina's transport restructuring through the lens of market access. We construct a unified market access index that integrates both rail and road networks using a standard spatial economics framework, computing generalized transport costs and shortest-path routes on the historical networks. Our main analysis estimates how regional outcomes—population, sectoral economic activity, and additional outcomes including education and employment—respond to changes in total market access induced by the observed infrastructure plan. We then use counterfactual market access measures to explore the relative roles of rail closures versus road expansion, constructing separate scenarios where only rail changes (roads frozen) or only roads change (rail frozen).

Our empirical framework follows the market access approach standard in spatial economics and trade. We model each district's economic outcomes as functions of its market access, defined as the population-weighted sum of inverse generalized transport costs to all other districts. To address endogeneity, we instrument for observed market access changes using two sources of variation. First, we exploit the Larkin Plan's technical rules: the World Bank study that motivated the reforms only examined railway segments below a specific freight threshold for potential closure, so studied segments faced a much higher closure probability than nonstudied ones. Second, we construct hypothetical road networks connecting major cities via least-cost paths, which predict road expansion but are orthogonal to local economic shocks.

We construct a novel dataset combining historical transport networks with census data at the district level, covering 312 districts across Argentina. We validate our identification strategy with pre-period balance checks and a pre-trends placebo, examine sensitivity to spatially correlated standard errors in Section 8, and report where the validation package falls short: the pre-trends placebo rejects a clean null at the five-percent level in the OLS and combined-IV specifications, a rejection that spatially robust standard errors do not soften, and we state that limitation rather than obscure it.

The counterfactual decomposition qualifies the modal question that motivates the paper. Recomputing market access with one network frozen at its pre-reform configuration, we find that the only-rail and only-road changes contribute similar point estimates to the population response (0.036 and 0.050, against 0.052 for the full change), while differing sharply in identification: the rail component, instrumented by the Larkin Plan alone, is the best-identified object in the paper, with a first-stage F above 100 for total population. We find no support for the view that one mode's market-access channel dominates the other; the evidence is instead consistent with population responding to market access itself, whichever mode produces it.

The mechanism decomposition shows where that response lives. Adding own-district infrastructure changes—rail and road kilometers and two binary margins—to the main specification removes roughly half of the market-access coefficient (0.052 to 0.029), so about half of the estimated response operates through infrastructure located in the district itself and the remainder through regional connectivity beyond it. Suggestive heterogeneity patterns point in the same direction as theory: the market-access coefficient is larger in initially smaller, more rural, and more remote districts, although these interactions are too imprecisely identified to support magnitude claims.

Our main results document a contrast between population and sectoral activity. The elasticity of total population with respect to $\Delta \ln \mathrm{MA}^{\mathrm{full}}$ is 0.052 (0.031) in the combined-IV specification, significant at the ten-percent but not the five-percent level; the corresponding OLS estimate is significant at the five-percent level, and the urban and rural coefficients are positive but less precisely estimated. The evidence for an aggregate population response is suggestive rather than conclusive. In contrast, the elasticity of manufacturing activity is sizable and significant: the log change in the value of manufacturing production has an elasticity of 0.317 (0.120) and the log change in the manufacturing wage mass has an elasticity of 0.378 (0.124). Agricultural activity, measured by the number of farms and by total farmed area, does not respond detectably. The contrast survives measuring each sector's market access under a cost schedule matched to its cargo profile, and strengthens for manufacturing. This pattern is consistent with the scale-economies hypothesis introduced in Section 2: the transport restructuring shifted the cost structure faced by different sectors differently, with manufacturing benefiting more than agriculture. Robustness to alternative trade elasticities, alternative hypothetical-road instruments, and subsample choice is reported in Section 5.5; the level of the population elasticity, though not the sectoral contrast, depends on the trade-elasticity calibration, a dependence Section 8 quantifies.

This paper contributes to several literatures. First, we provide causal evidence on the regional effects of infrastructure disinvestment driven by fiscal consolidation. The Argentine episode involved not just the removal of rail infrastructure but a technological shift in the dominant transport mode—from rail to road—that effectively acted as an unintentional industrial policy, reshaping the country's economic geography. Second, we contribute to understanding mode-specific effects in multimodal infrastructure episodes, proposing a counterfactual market access approach that clarifies what can and cannot be identified. Recent structural work models multimodal freight networks with explicit mode choice and terminal congestion for the contemporary United States and finds that abstracting from multimodal flexibility can understate the welfare gains from infrastructure improvements (Fuchs and Wong, 2026); we provide the causal, historical complement, estimating the regional effects of a multimodal restructuring that actually occurred, with quasi-experimental identification. Third, our findings speak to contemporary policy debates about modal shifts in transport, as many countries face ongoing transitions between rail and road. Fourth, we contribute to a longstanding debate in Argentine economic history about whether railroad closures caused the decline of hinterland towns or merely followed pre-existing trends.

The paper proceeds as follows. Section 2 provides historical context. Section 3 describes data and market access construction. Section 4 presents the empirical strategy. Section 5 reports main results. Section 6 explores mode-specific patterns. Section 7 examines mechanisms and heterogeneity. Section 8 interprets the results and states their limitations, and Section 9 concludes.

2 Historical Context

2.1 Argentina's Railway System and Fiscal Crisis

Argentina's railway network was one of the largest in the world. Construction began in 1857 and expanded rapidly through a combination of national capital, foreign investment—predominantly British and French—and direct government participation extending lines into rural areas that were not commercially profitable. By the early twentieth century, the network exceeded 40,000 kilometers, designed primarily to connect the agricultural hinterland of the Pampas, the Northwest, and the Northeast to the port of Buenos Aires. This radial, export-oriented design was both cause and consequence of the agricultural export boom of the first globalization: from 1869 to 1914, Argentine real exports grew approximately 500 percent (Fajgelbaum and Redding, 2022). By the start of the First World War, Argentina was the eighth richest country in the world.

Starting in the 1930s, export growth slowed considerably and profit margins of railroad companies declined. Investment in railroad infrastructure stalled, and the aging network experienced rising operational costs. Railroads also began to face competition from the automotive industry—Ford produced the first Model T in Latin America in Buenos Aires in 1925. In 1932, Law 11.658 created a National Trunk Road System and a lump-sum fuel tax whose proceeds were allocated entirely to road construction and improvement; by 1935, there were almost 3,000 kilometers of paved roads. During the Second World War, Argentina remained neutral and its young industrial sector flourished while factories in the developed world were retooled for war production. Railroads experienced a "second life" as wartime food shortages drove up international prices for Argentine agricultural exports, helping the country accumulate reserves of $1.6 billion (Whitaker, 1954). But by 1945, industry's contribution to GDP surpassed that of agriculture for the first time (Potash, 1969).

Supported by unions, the military, and the new industrial bourgeoisie, General Juan Domingo Perón reached the presidency in 1946. His economic policy rested on three pillars: strengthening the domestic market, centralized planning with state ownership of strategic sectors, and industrial development (Whitaker, 1954; Pahowka, 2005). The showcase of this program was the acquisition and nationalization of the private railroad companies in 1947 for $600 million—a purchase widely debated at the time, with economists including Raúl Prebisch arguing that the price was excessive given the crumbling state of the infrastructure (Rock, 1987). By the early 1950s, when the fortuitous advantages of the war years had faded, Argentina's economy slipped into depression. Perón moved to protect nascent industry, armoring the automotive and petrochemical sectors against international competition. The road network continued to grow at a slow, steady pace, reaching 9,000 kilometers of paved roads by 1955. The railroad network, meanwhile, remained stable at around 43,500 kilometers, but with a shrinking budget it suffered from poor maintenance and aging equipment.

In 1955, a coup removed Perón from government, and the country endured several years of political and economic instability. In 1958, Arturo Frondizi was elected president in an election from which Perón was banned. Frondizi's economic plan centered on fostering industrialization through foreign direct investment: Decree 3693 of 1959 promoted investment in the national automotive industry, and 23 new factories were approved within a year. His administration was equally aggressive in promoting road construction. In 1958, the lump-sum fuel tax was replaced by a 35 percent ad valorem tax (Decreto Nacional 505/1958), and in 1960 Law 15.274 instituted a sales tax on vehicles—in both cases, funds were permanently allocated to road construction and improvement. As for railroads, Frondizi initially sought to modernize the crumbling infrastructure, but the investment required to achieve a modern network while maintaining its current size was nearly prohibitive (López and Waddell, 2007). By 1960, annual railroad losses had reached approximately $280 million per year, accounting for almost 80 percent of the federal government deficit (Keeling, 1993).

2.2 The Larkin Plan (1960–1962)

In this context, the Argentine government agreed with the World Bank to commission a team of experts to evaluate the long-term economic and strategic viability of the national transport infrastructure. The team was led by General Thomas Larkin, an American expert in military logistics who had been decorated for his efforts supplying combat troops during the Second World War and had experience consulting on railroad modernization in France, Germany, and Japan. Between 1959 and 1961, the World Bank mission scrutinized the profitability of the existing railroad and road networks. The resulting study, published by the Ministry of Public Works and Services in 1962 as *Transportes Argentinos: Plan de Largo Alcance* ("Argentine Transportation: Long-Run Plan"), comprised three volumes totaling approximately 3,000 pages.[1] The first volume contained an executive summary with diagnosis, methodology, results, and recommendations organized by transport mode and time horizon; the remaining two volumes, of approximately 1,000 pages each, provided detailed analysis.[2]

In a world of cheap and stable oil prices, and given the poor material and financial condition of the railroad system, the plan's main recommendations were dramatic: closure of approximately 15,000 kilometers of railroad track (roughly 32 percent of the existing network), dismissal of around 70,000 railroad employees (from a total of approximately 200,000), disposal of the entire fleet of steam vehicles, closure of most national railroad workshops and factories, and aggressive improvement and construction of paved roads—in some cases to substitute for closed railroad segments, but also to create new connections in the network (Larkin, 1962). The plan proposed the construction of approximately 10,000 kilometers of paved roads over ten years.

A key feature of the Larkin Plan's design is that it did not study the entire railroad network with equal intensity. Due to budget constraints and the vast size of the existing network, only railroad segments that fell below specific freight density thresholds were "studied" in detail—meaning that the experts checked each segment's explicit revenues and costs and analyzed whether closure was warranted. The thresholds were 1 million gross tons per year of cargo passing through a given segment, or 500 tons per kilometer per year of cargo entering the network through a given segment. By the report's own accounting, only 39.6 percent of the railroad network was studied. For each studied segment, the experts issued one of three recommendations: maintain, close, or perform a new study. Crucially, the likelihood of the plan recommending closure was much higher for studied segments than for nonstudied ones—a classification that we exploit for identification.

The plan's recommendations were announced by President Frondizi in 1961, but massive protests and strikes erupted in many cities. After several months of clashes and social unrest, the government decided to abandon the plan. In the short term, only approximately 1,000 kilometers of track were closed. However, despite being abandoned after just a few months, the Larkin Plan was the only

[1] The report's cover and its letter of elevation to the minister are dated 1962; President Frondizi announced the plan's recommendations in 1961, before the full report was issued.

[2] We accessed physical copies of the plan at the Mariano Moreno National Library in Buenos Aires. We scanned and digitized two key objects: a map describing the operational railroad network in 1960 (our baseline) and three tables describing the diagnosis and recommendations for each railroad segment.

long-term comprehensive study of the national transport infrastructure for more than 30 years. It is therefore likely that its recommendations influenced national transport policy for decades ahead. As we document below, railroad closures in the 1960s and 1970s were approximately three times more likely for segments that had been studied by the plan, probably because no new large-scale study was ever undertaken to supersede it.

2.3 Implementation: Rail Closures and Road Expansion, 1960–1991

The Larkin Plan marked a trend break in the evolution of both transport networks (Figure 1). For railroads, it initiated a long-term disinvestment and dismantling process. For roads, it started an almost two-decade period of aggressive construction and improvement. These large-scale changes occurred in a context of population growth (from 20 to 33 million between 1960 and 1991) and increasing GDP per capita (growing approximately 35 percent at an average annual rate of 0.95 percent).

Railroad closures proceeded in two distinct waves. The first wave, from 1960 to 1966, closed approximately 4,000 kilometers of track before political opposition brought the process to a halt. The second wave came under the military dictatorship that seized power in 1976: without union resistance, the military government ordered the immediate closure of over 6,000 additional kilometers of track between 1976 and 1979. After 1979, the network stabilized—no further closures or new construction occurred between 1979 and 1986. In total, the railroad network fell from approximately 43,500 kilometers in 1960 to approximately 33,500 kilometers by 1986, a reduction of roughly 23 percent.[3]

Road expansion, by contrast, was continuous throughout the period. Fueled by the dedicated fuel and vehicle taxes established under Frondizi, the paved road network grew from approximately 10,000 kilometers in 1954 to approximately 28,000 kilometers by 1986—an increase of roughly 180 percent. Gravel roads also expanded, though more modestly. By 1986, the combined paved and gravel road network had reached approximately 35,000 kilometers, roughly equal to the remaining railroad network.[4] Importantly, the spatial pattern of road expansion did not simply mirror the pattern of railroad closures: roads were not systematically built to replace closed rail lines. As we show in Section 3, the correlation between rail closures and road expansion at the district level is weak, meaning that the two shocks affected largely different sets of districts.

[3] We end our sample period in 1991 because there was a census in that year and because we wish to abstract from the potentially different effects of the privatizations of large fractions of the railroad and road networks during the 1990s.

[4] Figures in this subsection come from the Dirección Nacional de Vialidad series for the national road network (Dirección Nacional de Vialidad, n.d.): 27,276 km paved plus 7,153 km gravel in 1986, excluding dirt roads. The digitized road maps used in the empirical analysis (Section 3) cover the broader network, including provincial roads, and therefore report larger totals.

2.4 Scale Economies and Sector-Mode Specialization: Motivation and Hypotheses

A feature of the Argentine setting that motivates part of our analysis is that railroads and roads may serve different economic sectors, owing to fundamental differences in their cost structures. Railroads exhibit strong scale economies: high fixed costs (track maintenance, stations, rolling stock) but low variable costs per ton-kilometer at high traffic density. Roads, by contrast, have more constant returns to scale—per-unit costs decline less steeply with volume. The Larkin Plan itself was designed around this logic: it studied segments precisely because their traffic density was too low for rail's scale economies to be realized.

Data from Baumgartner and Palazzo (1969), who estimated transport costs for Argentina using detailed operational data, are consistent with this pattern. At high cargo density (1,000 tons per day), railroad costs were 0.465 pesos per ton-kilometer compared to 1.000 for roads—rail was more than twice as cheap. At low cargo density (100 tons per day), the relationship reversed: railroad costs rose to 13.490 pesos per ton-kilometer while road costs were 8.266—road was substantially cheaper (Table 2). If these cost differences translated into actual shipping patterns, they would imply a natural sector-mode specialization: agricultural products—bulky, heavy, shipped in large volumes—would fall in the high-density regime where rail has a cost advantage, while manufactured goods—lighter, more varied, shipped in smaller batches—would fall in the low-density regime where road is cheaper.

The cost structure is not the only conceptual difference between the two shocks. A second difference is the margin of change: the road network grew through construction and the paving of existing routes, while the rail network contracted through the closure of existing lines. The two margins need not have symmetric local effects. A closure removes a service around which firms, storage facilities, and settlements had already located, so its effects would operate through the depreciation of place-specific investments; a new or newly paved road adds access whose complementary investments are still partly to come, so its effects would operate through entry and upgrading. If adjustment to losses is slower than adjustment to gains—structures and settled populations are durable—the same absolute change in access can produce responses that differ with its sign.

A third difference is network geometry. The railroad network was laid out radially, oriented to move export commodities from the interior to the port of Buenos Aires (Section 2.1), so many interior region pairs were linked mainly through that hub; the road network is visibly more capillary, connecting interior districts to one another as well as to the port (Figure 1). A contraction of the radial network combined with an expansion of the capillary one therefore changes not only the level of a district's market access but also its composition—which destinations become cheaper to reach—in ways a single aggregate index summarizes only partially.

Whether these differences—in cost structure (the density channel developed above), in the margin of change, and in network geometry—generate meaningful mode-specific effects on regional economic outcomes is ultimately an empirical question. It is not obvious a priori that they do: if road expansion sufficiently compensated for rail closures, or if firms and workers adjusted quickly to the new transport configuration, the sectoral composition of economic activity might not have changed. We treat the existence of mode-specific effects as a hypothesis to be tested in Sections 5 and 6, rather than as an established fact. The cost data presented here motivate the density channel directly; the margin-of-change and geometry channels are stated as context for interpreting the mode-specific estimates, and our district-level data do not separate the three channels from one another.

3 Data

The empirical analysis combines three data components. Section 3.1 describes the transport network geometries and their changes between 1960 and 1986. Section 3.2 describes the district-level population and economic outcomes. Section 3.3 describes how we construct market access from transport cost surfaces.

3.1 Transport networks

We combine two georeferenced networks: the railway network at its 1979 status, which we map to 1960 and 1986 via documented closure dates, and the paved-and-gravel road network at three cross sections (1954, 1970, 1986).

Railways. The rail network comes from the network map in Argentina's 1979 *Política Ferroviaria* report (Secretaría de Estado de Transporte y Obras Públicas, 1981), which we digitised against official IGN geographic references.[5] Each digitised segment carries the information needed to reconstruct the network and to build the instrument. First, its operating history: whether the segment remained active through 1986, was closed between 1960 and 1966, or was closed during the military dictatorship (1976–1983). We use these closure dates to reconstruct the network as it stood in 1960 and as it stood in 1986. Second, its Larkin Plan status: whether the segment was among the 237 segments the 1962 Larkin Plan selected for detailed study — segments falling below the Plan's freight-density thresholds, whose closure it then evaluated on explicit revenue and operating-cost grounds — or among the 328 segments the Plan did not evaluate. For studied segments, the digitisation also records which of the Plan's three recommendations (maintain, close, or perform a new study) was issued. The studied/non-studied distinction is the basis of the first instrument (Section 4).

Total rail kilometres fall from 42,780 in 1960 to 33,303 in 1986, a 22% reduction. Of the 9,477 km lost, 5,563 km (59%) were closed during the dictatorship and 3,914 km (41%) were closed between 1960 and 1966. Figure 1a shows the geography of the rail network and highlights the dictatorship closures in red. 30 of the 312 districts have no rail in any period; these are concentrated in Patagonia and Tierra del Fuego.

[5] The report's Figure V–5 was scanned, georeferenced, and hand-traced in GIS; the resulting file contains 565 line segments. District-level rail kilometres are computed by intersecting the segments with the district polygons.

Roads. The road network comes from a digitised comparison of Automóvil Club Argentino road maps for 1954, 1970, and 1986 (Automóvil Club Argentino, 1954–1986). Each of the 1,741 digitised segments is classified by the set of map years in which it appears, which lets us reconstruct the paved-and-gravel network at each of the three cross sections.[6] The network grows from 27,513 km in 1954 to 79,820 km in 1986, a 190% increase. Figure 1b shows the road network and highlights the segments added between 1954 and 1986 in red. Table 1 summarises the network totals by period.

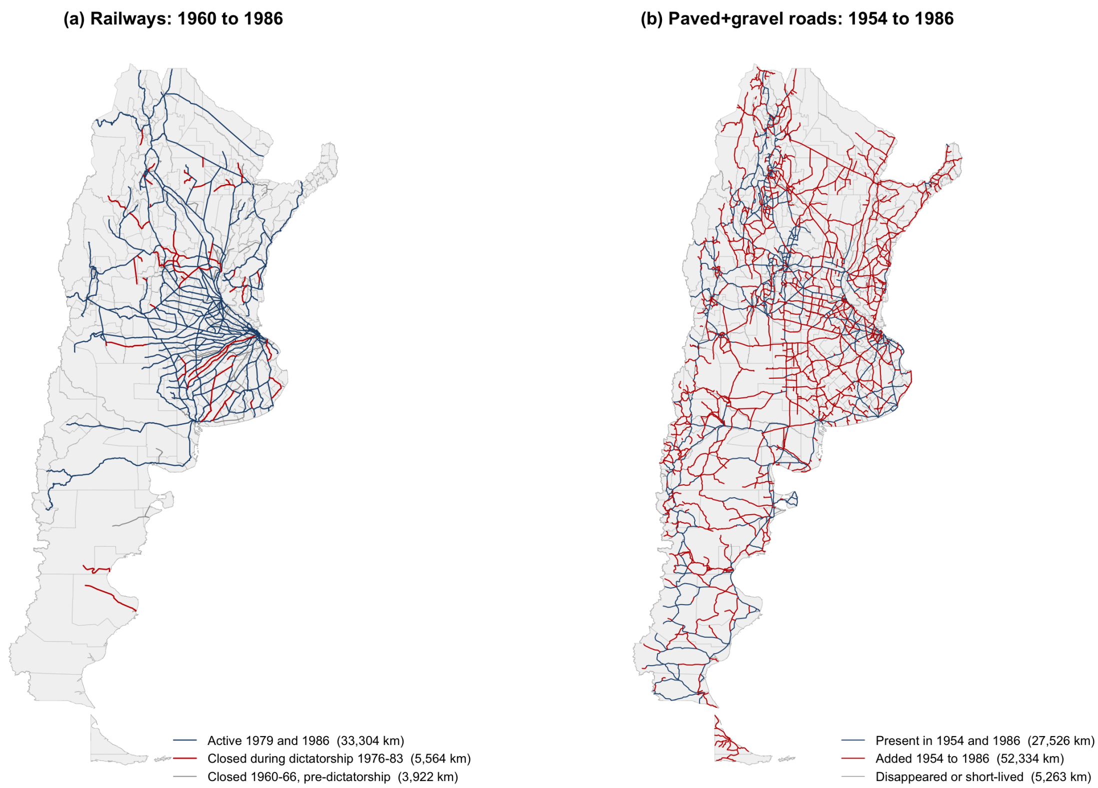

Figure 1: Transport network changes, 1960–1986.

Three districts, all in Patagonia, lack any segment in the road shapefile and contribute missing values in the district-level road variables. 11 districts lost rail without gaining roads and 141 districts both lost rail and gained roads. The cross-district correlation between Δ(rail km) and Δ(road km) is -0.25.

Hypothetical road networks. We also construct four hypothetical road networks that connect Argentina's largest cities under stylised routing rules. The node set is the 54 cities that were provincial capitals or had population of at least 15,000 in 1960. The first network links every pair of cities with a straight line (1,431 segments); the second is the minimum spanning tree of those straight-line distances (53 edges). The third replaces straight lines with least-cost paths routed over a construction-cost surface that, following Faber (2014), penalises slope, wetlands, urban land, and

[6] Segments that appear on the 1954 and 1986 maps but not on the 1970 map are treated as continuously present (a cartographic omission on the middle map). Segments absent from the 1986 map are excluded from the period totals.

Table 1: Transport Network Changes, 1960–1986

	1960	1986
Panel A. Railways		
Total km	42,780	33,303
Net change		-9,477 (-22.2%)
closed during dictatorship 1976–83		5,563
closed 1960–66, pre-dictatorship		3,914
Panel B. Paved+gravel roads		
	1954	1986
Total km	27,513	79,820
Net change		52,307 (+190.1%)
1970 total		37,925

water crossings; the fourth is the minimum spanning tree of those least-cost-path weights. These networks serve as instruments for the actual road network in Section 4; the least-cost-path spanning tree is the main choice, and the other three appear as robustness alternatives.

3.2 Census data and outcomes

Geographic unit. We work at the district (*departamento*) level using IPUMS International's time-invariant district geography for Argentina (Minnesota Population Center, 2020), which partitions the mainland into 312 districts, including Capital Federal and the two Tierra del Fuego jurisdictions. The same polygons apply across the five IPUMS sample years (1970, 1980, 1991, 2001, 2010), the 1947 and 1960 census publications, and the agricultural and industrial censuses, so every outcome is measured on identical boundaries throughout.

Population. Population in 1960 comes from the digitised *Censo Nacional de Población 1960* tables (Dirección Nacional de Estadística y Censos, 1960b), reconciled to the IPUMS district geography through a crosswalk that handles pre-1960 splits and name changes. The mean 1960 district population is 52,965 with a national total of 16.5 million. Population in 1991 comes from the IPUMS microdata aggregated to the same district geography; the mean 1991 district population is 114,723 with a national total of 35.8 million. Log population change $\Delta \log(\text{pop}_{91/60})$ has mean +1.02 and standard deviation 0.54. Capital Federal and the two Tierra del Fuego districts are covered by both census years; Capital Federal contributes to the market-access population weights defined in Section 3.3 but is excluded from the estimation sample as an observation (Section 5).

Urban population. Urban population shares at the district level come from the 1960 census (classification per INDEC) and from the IPUMS 1991 microdata. Urban population is not coded consistently in the 1970 and 2010 IPUMS microdata for Argentina; those years are recorded as missing in the panel and are not used for urban-outcome regressions. 1960 and 1991 give us the post-reform window with consistent urban/rural classification.

Pre-reform placebo. We supplement the baseline with the 1947 national census (Dirección Nacional del Servicio Estadístico, 1947), which covers 243 of the 312 districts. A log population change $\Delta \log(\text{pop}_{60/47})$ for this subsample serves as a placebo outcome: it measures population growth before any of the 1960–1986 network changes, and should be uncorrelated with the instruments if the instruments isolate post-1960 network-induced variation.

Industrial and agricultural outcomes. Sectoral outcomes come from four additional censuses. The 1954 and 1985 industrial censuses (*Censo Industrial* and *Censo Económico*; Dirección Nacional de Estadística y Censos 1954; Instituto Nacional de Estadística y Censos 1985) provide district-level establishment counts, total wages paid (the wage mass), and production values, each measured on definitions comparable across the two census years. Employment is recorded on incompatible definitions in the two years (workers in 1954, total persons engaged in 1985) and is not used. The 1960 and 1988 agricultural censuses (Dirección Nacional de Estadística y Censos, 1960a; Instituto Nacional de Estadística y Censos, 1988) provide the number of farm operations and total agricultural area. All monetary variables are nominal pesos of their census year; real comparisons across years require a deflator that we defer to robustness.

Geographic controls. District-level controls come from external sources. Area in km^2 is computed from the IPUMS polygon. Mean elevation comes from GTOPO30 (U.S. Geological Survey, 1996). Terrain ruggedness uses the Nunn–Puga TRI raster (Nunn and Puga, 2012) derived from GTOPO30. Wheat potential yield comes from FAO–GAEZ 3.0 (FAO and IIASA, 2012). Caloric potential (pre- and post-1500) comes from Galor and Özak (2016). Distance to Buenos Aires is the great-circle distance from the centroid of the Gran Buenos Aires polygon in the IGN settlements shapefile (Instituto Geográfico Nacional, 2026b). All continuous controls are standardised (zero mean, unit variance) for use in the regressions. Appendix Table A1 reports descriptive statistics for every variable entering the estimation.

3.3 Market access

Definition. Following Donaldson and Hornbeck (2016), we define district i's market access in period t for sector s under elasticity θ as:

$$\text{MA}_{i,t,s,\theta} = \sum_{j \neq i} \frac{\text{pop}_{j,1960}}{(\tau_{ij,t,s})^{\theta}}, \tag{1}$$

where $\tau_{ij,t,s}$ is the origin–destination transport cost between the centroids of districts i and j under network configuration t and sector s. Population weights are held fixed at their 1960 values across all t so that variation in $\text{MA}_{i,t,s,\theta}$ comes entirely from variation in transport costs rather than from endogenous population responses to the network changes. All 312 districts contribute to the weights, including Capital Federal (2.97 million people in 1960): access to Buenos Aires enters every district's market access even though Capital Federal itself is not an observation in the estimation sample (Section 5).

Cost surface. Transport costs $\tau_{ij,t,s}$ are the least-cost-path distances on a 2,399 × 3,090 raster covering mainland Argentina in the ESRI:54034 equal-area projection (approximately 1.2 km per cell). The per-pixel cost $c_i(t,s)$ is:

$$c_i(t,s) = \begin{cases} \text{cost}_{\text{nav}}(s) & \text{if pixel } i \text{ is navigable water,} \\ \text{cost}_{\text{mininfra}}(s) & \text{if pixel } i \text{ has both rail and road,} \\ \text{cost}_{\text{rail}}(s) & \text{if pixel } i \text{ has rail only,} \\ \text{cost}_{\text{road}}(s) & \text{if pixel } i \text{ has road only,} \\ \text{cost}_{\text{land}} \cdot \text{HMI}_i & \text{if pixel } i \text{ is off-network land in Argentina,} \\ \text{NA} & \text{otherwise (hard barrier).} \end{cases}$$

Network layers (rail, road, and navigation) are rasterised from the respective shapefiles after a 1 km buffer to guarantee at least one full raster cell per linear segment; the navigation layer comprises the watercourse segments flagged navigable in the IGN hydrography shapefile (Instituto Geográfico Nacional, 2026a). The Strait of Magellan is buffered by 5 km so that at least one raster cell per shore becomes navigable, which lets Dijkstra cross the strait and keeps Tierra del Fuego connected to the mainland. Off-network pixels outside Argentina are set to NA, which acts as a hard barrier for the least-cost path algorithm except where a navigable-water pixel overrides the country mask (e.g. the Paraná river border). Figure A4 maps the navigation layer: the navigable set is the inland Paraná–Paraguay–Uruguay–Río de la Plata system plus a few Patagonian rivers, together with the Magellan crossing. The cost surface contains no open-ocean coastal shipping, so Atlantic ports enter the network through land or river legs only.

Cost parameters. Sector-specific cost parameters come from Baumgartner and Palazzo (1969) Table II. Three sectors $s \in \{0,1,2\}$ correspond to three cargo density scenarios (medium, high, low); Table 2 summarises the values.

In particular, rail is cheaper than road at high cargo density ($\text{cost}_{\text{rail}}(s{=}1) = 0.465 < \text{cost}_{\text{road}}(s{=}1) = 1.000$) but more expensive at low density ($\text{cost}_{\text{rail}}(s{=}2) = 13.490 > \text{cost}_{\text{road}}(s{=}2) = 8.266$), which is the scale-economies channel that motivates sector-specific analysis. At medium density, rail and road costs are nearly identical (1.874 vs. 1.777), so mode choice matters less for the overall market-access index. We therefore use the medium-density scenario as the main specification — it is the general-purpose schedule and does not build a mode preference into the headline treatment — and report the population estimates under all three schedules alongside the main results (Table 10). The off-network cost $\text{cost}_{\text{land}}$ is $17.9 \times \text{cost}_{\text{road}}(s{=}2)/\overline{\text{HMI}}$, where $\overline{\text{HMI}} = 0.2018$ is the national mean of the Özak (2018) Human Mobility Index. The factor 17.9 is the ratio of motor-vehicle

Table 2: Transport Costs by Mode and Cargo Density (1960 pesos per ton-km)

	Low density (100 t/day)	Medium density (500 t/day)	High density (1,000 t/day)
Road	8.266	1.777	1.000
Railroad	13.490	1.874	0.465
Navigation	0.621	0.621	0.621

Notes: Ground-transport costs from Baumgartner and Palazzo (1969). Navigation costs from Larkin (1962). Medium-density costs are used in the main analysis (sector $s = 0$); the headline population result is re-estimated under the high- and low-density schedules (sectors $s = 1$ and $s = 2$) in Table 10.

speed to walking-equivalent speed, calibrated on a Buenos Aires–Rosario comparison. This gives $\text{cost}_{\text{land}} \approx 733.2$; the per-pixel off-network cost scales with the local HMI value and is one to two orders of magnitude above on-network cost.

Least-cost path and elasticity. We compute $\tau_{ij,t,s}$ by running Dijkstra's algorithm over the cost surface between the 312 district centroids, separately for each network configuration and sector.[7] $\tau_{ii,t,s} = 0$ by construction and τ is symmetric.

The elasticity θ controls how sharply distant destinations are discounted in the MA sum. It is taken from the trade literature, not estimated in this paper. Structural estimates span a wide range: Eaton and Kortum (2002) report a preferred value of 8.28; Simonovska and Waugh (2014) revise the Eaton–Kortum approach with simulated-moments estimation and obtain values near 4, with a credible range of roughly 4 to 8; Caliendo and Parro (2015) estimate 8.11 for agriculture; and Donaldson (2018) estimates 3.8, the only estimate from railroad-borne trade (colonial India). We report two values within this range: $\theta_{\text{low}} = 4.55$, a midpoint choice within the modern range of estimates rather than a value taken from any single study — the Simonovska and Waugh (2014) benchmark of approximately 4.1 and the Donaldson (2018) estimate sit nearby — and $\theta_{\text{high}} = 8.11$, the Caliendo and Parro (2015) agricultural trade elasticity. Main results use θ_{low}; θ_{high} appears in the sensitivity tables, and Section 5.5 reports the estimates over the full sweep $\theta \in [1, 12]$ (Table 21); Section 8 discusses what the sweep implies for comparisons with the literature. **[PLACEHOLDER:** whether a trade-elasticity exponent is the right object for our raw generalized cost τ is memo Decision A, unresolved**]**

Change in log MA. The main treatment variable is $\Delta \log \text{MA}_i \equiv \log \text{MA}_{i,1986,s=0,\theta_{\text{low}}} - \log \text{MA}_{i,1960,s=0,\theta_{\text{low}}}$. Across the 312 districts, $\Delta \log \text{MA}_i$ has mean +1.51, standard deviation 2.45, minimum −7.59 and maximum +9.53; 90.7% of districts see a positive change. Figure 2 maps $\Delta \log \text{MA}_i$ across districts.

Counterfactual decomposition. We additionally construct two counterfactual cost surfaces that freeze one network mode at its baseline while letting the other change:

[7] Paths move across an eight-connected grid; the computation uses the `gdistance` R package. The replication package documents runtimes.

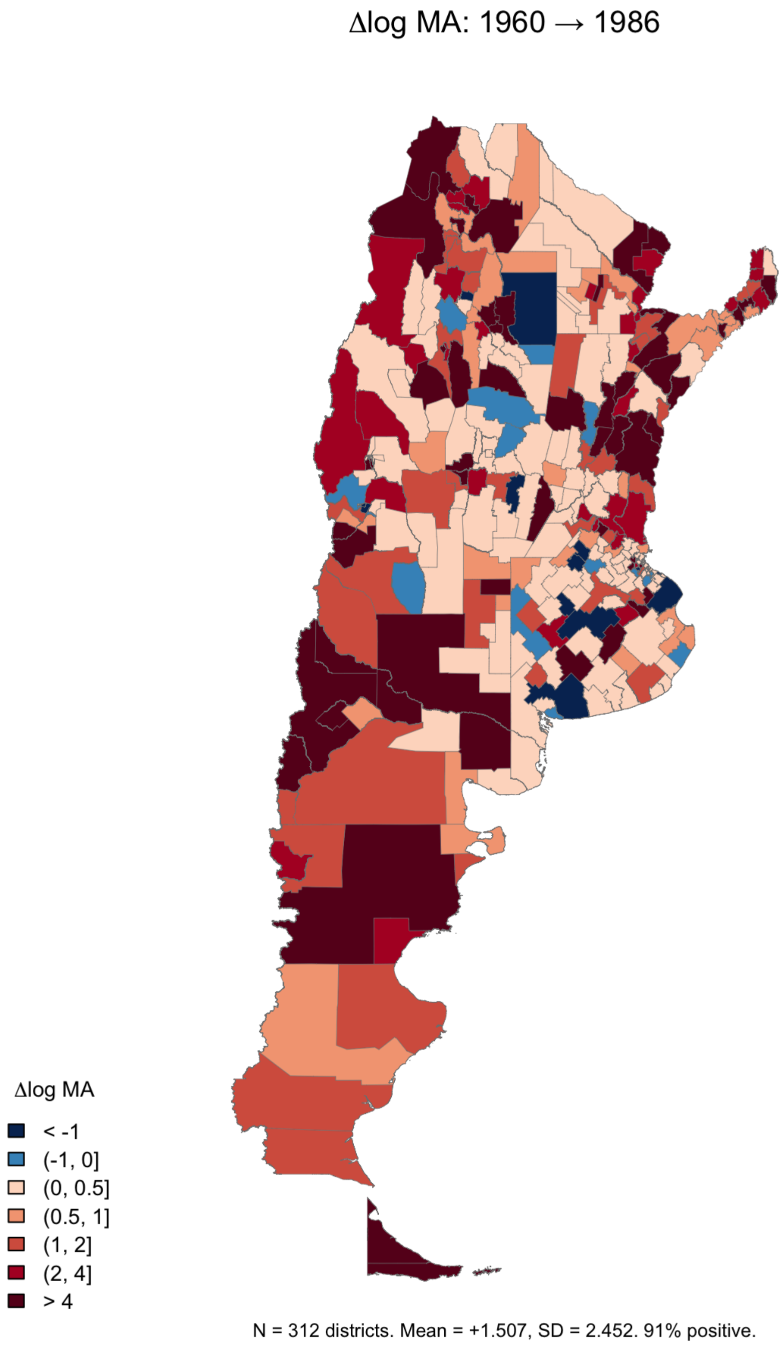

Figure 2: Change in log market access, 1960–1986.

- *only-rail*: 1986 rails combined with 1954 roads. $\Delta \log \mathrm{MA}_i^{\text{only-rail}} = \log \mathrm{MA}(\text{only-rail})_i - \log \mathrm{MA}_{i,1960}$.
- *only-road*: 1960 rails combined with 1986 roads. $\Delta \log \mathrm{MA}_i^{\text{only-road}}$ defined analogously.

At sector 0 and θ_{low}, $\Delta \log \mathrm{MA}^{\text{only-rail}}$ has mean -0.39 (rail contraction is a small market-access loss) and $\Delta \log \mathrm{MA}^{\text{only-road}}$ has mean $+1.69$ (road expansion is a large gain). The sum of the two isolated components ($+1.30$) is slightly below the total $\Delta \log \mathrm{MA} = +1.51$, reflecting a positive complementarity between the two network shocks. Figure 3 compares the three maps side by side.

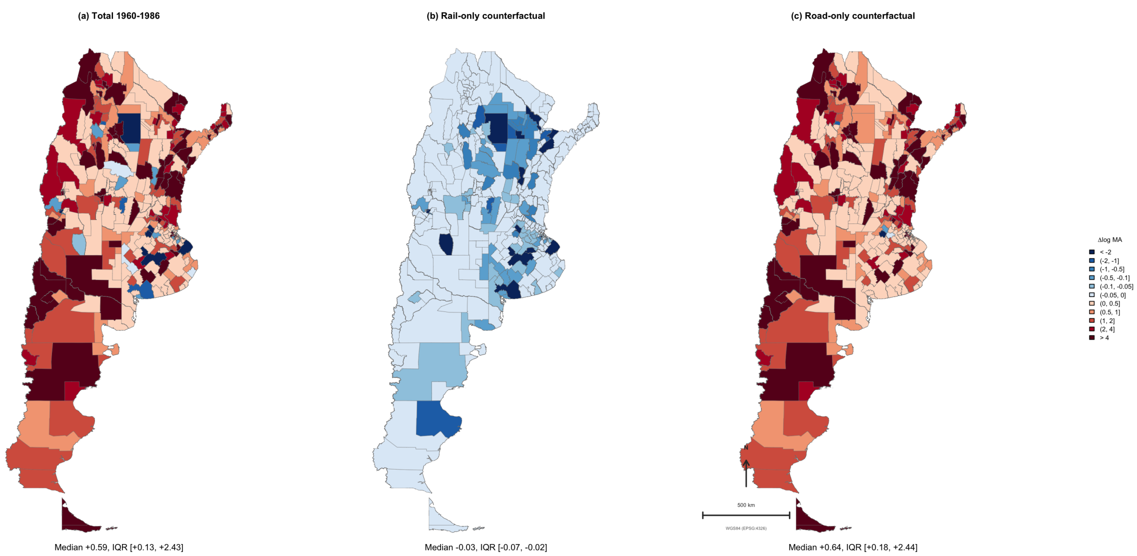

Figure 3: Market access changes under three scenarios: full, only-rail counterfactual, only-road counterfactual.

Limitations. The cost-surface construction imposes three modelling choices that the Section 4 discussion of identification returns to. First, mode switches are free: Dijkstra can step from a rail pixel to a road pixel at any grid boundary, whereas in reality such switches occur at stations or terminals and incur transshipment costs. Section 5.5 bounds this assumption from the opposite extreme (one mode per trip, i.e. infinite transshipment cost). Second, the algorithm selects a single cheapest path, not a probability distribution over routes; in equilibrium firms use multiple parallel routes. Assigning each cell the cheapest mode deterministically also treats modes as perfect substitutes along the path; Fuchs and Wong (2026) estimate an elasticity of modal substitution near 1.1 between rail and truck in the contemporary United States, far from perfect substitution. Third, cost parameters are constant across space and time; we do not model congestion or regional cost heterogeneity. Each of these simplifications follows the approach in Donaldson and Hornbeck (2016) and is common in the market-access literature.

4 Empirical Strategy

4.1 Estimating Equation

Our main specification relates changes in district-level outcomes to changes in market access:

$$\Delta Y_i = \beta \cdot \Delta \ln \mathrm{MA}_i^{\mathrm{full}} + X_i'\gamma + \varepsilon_i, \tag{2}$$

where ΔY_i is the change in outcome, $\Delta \ln \mathrm{MA}_i^{\mathrm{full}}$ is the change in log market access, and X_i includes baseline log market access, baseline log population, and geographic characteristics. The parameter β is the elasticity of regional outcomes with respect to market access.

4.2 Endogeneity Concerns

The regressor $\Delta \ln \mathrm{MA}_i^{\mathrm{full}}$ is not exogenous. The infrastructure changes that generate it — rail closures and road construction between 1960 and 1986 — were policy choices, not random assignments, and those choices plausibly responded to the same district conditions that drive the outcomes.

The direction of the resulting bias is ambiguous because infrastructure was allocated under two opposing logics. Under an efficiency logic, investment follows expected growth and divestment follows expected decline: roads are built, and rail retained, where outcomes were already rising, while rail is closed where outcomes were already falling. This aligns market-access gains with positive outcome shocks and losses with negative shocks — a positive correlation between $\Delta \ln \mathrm{MA}_i^{\mathrm{full}}$ and ε_i that biases the ordinary-least-squares estimate of β upward. Under a compensatory logic, investment instead targets lagging districts — for example, roads extended to areas that were declining — aligning market-access gains with negative outcome shocks and biasing β downward. Because both logics plausibly operated over 1960–1986, the sign of the net bias is ambiguous. We address the endogeneity with two instruments that move market access through variation plausibly orthogonal to district-level outcome shocks, described next.

4.3 Instrumental Variables

We instrument $\Delta \ln \mathrm{MA}_i^{\mathrm{full}}$ with two changes in market access, each computed on a counterfactual network: one that isolates rule-based rail closure, and one that isolates geography-driven road placement.

4.3.1 Instrument 1: The Larkin Plan Classification

The first instrument exploits the Larkin Plan (1960–1962), the government technical study of the rail network described in Section 2.2. The Plan evaluated segments against an engineering and freight-traffic standard and designated a subset for study toward closure; we label these segments *studied* and the remainder *non-studied*, mapped in Figure A3.[8] The classification predates and is

[8] In our cleaned data, studied segments account for 48.8 percent of network kilometers, compared with the 39.6 percent the report cites (Section 2.2). The two figures are consistent with a definitional difference. The plan's tables

defined independently of the regional-development outcomes we study. We construct $\Delta \ln \mathrm{MA}_i^{\mathrm{LP}}$ as the change in log market access between 1960 and 1986 on a counterfactual network in which the rail segments removed are exactly those the Larkin Plan studied, and we use it as an instrument for the realized change $\Delta \ln \mathrm{MA}_i^{\mathrm{full}}$.

The Larkin classification predicts realized closure: studied segments were substantially more likely to be closed than non-studied segments (Section 2.2), so the classification shifts the realized change $\Delta \ln \mathrm{MA}_i^{\mathrm{full}}$, which is the relevance condition the instrument must satisfy. The exclusion restriction requires that the classification be uncorrelated with district outcome shocks over 1960–1986, conditional on the controls in equation (2). The assumption is plausible because the Plan applied a national, traffic-based engineering rule fixed at the start of the period rather than a forecast of regional growth. It is not guaranteed: if the traffic rule selected segments in districts already on a distinct trajectory, studied status would correlate with pre-existing trends. Sections 4.4.1 and 4.4.2 probe this directly.

4.3.2 Instrument 2: Hypothetical Road Networks

The second instrument is a hypothetical road network that connects Argentina's major cities along the cheapest available routes, independent of where roads were actually built. The node set is the cities with at least 15,000 inhabitants in 1960 together with all provincial capitals. We connect these nodes in two ways. Edge cost is measured either as Euclidean distance or as the least-cost path across a terrain-based construction-cost surface, on which steeper and more rugged terrain is more expensive to traverse. From each cost notion we build a minimum spanning tree — the lowest-total-cost set of edges that connects all cities without cycles — which yields a parsimonious network that does not depend on which individual city pairs happen to be linked. This produces four variants: the full bilateral least-cost network (LCP) and its spanning tree (LCP-MST), and the full Euclidean network (EUC) and its spanning tree (EUC-MST); Figure A2 maps the variants. The main specification uses LCP-MST. We construct $\Delta \ln \mathrm{MA}_i^{\mathrm{hypo}}$ as the change in log market access induced by adding this hypothetical network.

The hypothetical network depends only on city locations and terrain, both fixed before the reform and unrelated to district outcome shocks over 1960–1986. It is relevant to the extent that actual road construction followed the same low-cost corridors connecting the same cities; this is an empirical relevance condition, evaluated by the first-stage fit rather than assumed. It is plausibly excludable because the least-cost routing reflects engineering geography fixed before the reform and does not respond to the local economic changes we study. As with the Larkin instrument, the exclusion restriction is assessed in the validation package (Sections 4.4.1 and 4.4.2).

assign every studied segment one of three recommendations (Section 2.2), and the segments recommended for *a new study* appear not to be counted as studied in the report's own accounting. In the raw digitized tables, where the studied share is 49.0 percent, the new-study segments account for 5,197 km; excluding them, the studied share is 38.4 percent of the approximately 43,500-kilometer 1960 network of Section 2.3. Our instrument retains these segments: they fell below the freight-density thresholds and were scrutinized by the plan, which is the classification the instrument exploits.

4.3.3 First Stage

The first stage regression is:

$$\Delta \ln \mathrm{MA}_i^{\text{full}} = \alpha_1 \cdot \Delta \ln \mathrm{MA}_i^{\text{LP}} + \alpha_2 \cdot \Delta \ln \mathrm{MA}_i^{\text{hypo}} + X_i'\delta + u_i. \tag{3}$$

4.4 Validation Package

Both instruments are formula instruments in the sense of Borusyak and Hull (2023): each is a market-access index computed from a counterfactual network, so each combines its source of identifying variation (the Larkin study designations; the least-cost-routing geography) with every district's non-random exposure to that variation through the population weights, district locations, and the remainder of the network. Borusyak and Hull (2023) show that instruments of this form can be correlated with outcome shocks even when the underlying variation is as good as random, because centrally located districts receive systematically different instrument values than peripheral ones. The validation exercises below address this concern directly: the balance table asks whether the instruments correlate with predetermined characteristics, the placebo asks whether they predict pre-period growth, and the main specification conditions on baseline log market access, baseline log population, and the geographic controls. These checks partially address, but do not eliminate, the exposure concern.

Sections 4.4.1 and 4.4.2 present two validation exercises that probe the exclusion restriction underlying the two instruments. Section 4.4.3 reports the first stage.

A specification with two instruments and one endogenous regressor imposes one testable restriction: both instruments must identify the same parameter. Sargan (1958) gives the classical test of that restriction. Standard errors throughout this paper are heteroskedasticity-robust, so we also compute the identification-robust counterpart, obtained by minimizing the Anderson and Rubin (1949) statistic over the parameter and referring the minimum to a chi-squared distribution with one degree of freedom. The same statistic delivers Anderson–Rubin confidence sets, which have correct coverage whatever the strength of the instruments; we report them where instrument strength is in question. Andrews et al. (2019) review both procedures. Rejection of the restriction does not identify which instrument is at fault. It says that the two-instrument estimate cannot be read as an estimate of a common parameter, and that the two instruments should be reported separately instead. Section 5.4 is organized around this test, and Table 17 reports it for the four outcomes considered there.

4.4.1 Pre-Period Balance

Table 3 reports regressions of each baseline district characteristic on the two instruments, partialling out baseline log market access, baseline log population, and the six geographic controls of the main specification; a characteristic that is itself one of these controls is excluded from its own control set. Each row is a separate regression of the row-label characteristic; columns report the instrument coefficients under three specifications: the Larkin Plan instrument alone (column 1), the hypothetical-road instrument alone (column 2), and both instruments jointly (column 3). Standard errors are heteroskedasticity-robust.

The Larkin Plan instrument is uncorrelated with baseline log population (column 1 coefficient -0.007, $p = 0.74$). It shows residual correlations with baseline urban share (0.013, $p = 0.01$), wheat suitability (-0.048, $p < 0.001$), pre-1500 caloric potential (-0.008, $p = 0.003$), post-1500 caloric potential (0.007, $p = 0.002$), distance to Buenos Aires (-0.043, $p < 0.001$), and log district area (-0.145, $p < 0.001$). These correlations are small in magnitude but statistically detectable. Wheat suitability, the two caloric potentials, and distance to Buenos Aires are controls in the main specification, so their residual correlations shift from exclusion-restriction threats to a conditional-independence requirement: the Larkin Plan instrument must be uncorrelated with the outcome *conditional on* these geographic controls.

The hypothetical-road instrument shows imbalance at the five-percent level for 2 of the nine baseline characteristics (baseline log population and log district area); the remaining column 2 coefficients show no correlation at conventional levels. Baseline log population is a control in the main specification. The joint specification in column 3 yields similar results to the single-instrument columns.

Table 3: Pre-period balance on the two instruments

	(1)	(2)	(3) Both	
Outcome	LP only	Hypo only	LP	Hypo
Log pop, 1960	-0.007	0.054**	-0.003	0.053**
	(0.023)	(0.024)	(0.023)	(0.024)
Urban share, 1960	0.013**	0.010	0.014***	0.011
	(0.005)	(0.007)	(0.005)	(0.007)
Log area (km^2)	-0.145***	-0.070***	-0.151***	-0.082***
	(0.025)	(0.026)	(0.024)	(0.026)
Elevation (std)	0.023*	-0.026*	0.021	-0.024
	(0.013)	(0.015)	(0.013)	(0.015)
Ruggedness (std)	0.009	-0.021	0.007	-0.021
	(0.013)	(0.017)	(0.013)	(0.017)
Wheat suitability (std)	-0.048***	-0.014	-0.050***	-0.017
	(0.012)	(0.014)	(0.012)	(0.014)
Caloric pot. pre-1500 (std)	-0.008***	0.003	-0.008***	0.002
	(0.003)	(0.003)	(0.003)	(0.003)
Caloric pot. post-1500 (std)	0.007***	-0.002	0.007***	-0.001
	(0.002)	(0.003)	(0.002)	(0.003)
Distance to B.A. (std)	-0.043***	-0.014	-0.044***	-0.017
	(0.011)	(0.017)	(0.011)	(0.017)
Observations	311	311	311	

Notes: Each row is one regression. The dependent variable is the row label; the regressors are the instrument(s) listed in the column headers plus baseline log MA (1960), baseline log pop (1960), and the six geographic controls of the main specification as partialled-out controls. When the row's characteristic is itself one of these controls, it is excluded from the control set rather than partialled out against itself. Robust (HC1) SE in parentheses.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

4.4.2 Pre-Trends

Table 4 reports a placebo test. The dependent variable is the log change in district population between 1947 and 1960 — a change that predates the 1960–1986 infrastructure restructuring. The regressor is $\Delta \ln \mathrm{MA}_i^{\mathrm{full}}$ computed over 1960–1986. If the infrastructure changes in 1960–1986 are driven by instruments orthogonal to pre-period district-level dynamics, the coefficient on post-reform $\Delta \ln \mathrm{MA}_i^{\mathrm{full}}$ in this placebo regression should be near zero.

The sample is 237 districts — the subset for which the 1947 census provides comparable population counts under the time-invariant district boundaries, as discussed in Section 3. Columns (1) to (4) match the structure of Tables 6 and 12: OLS, IV-LP, IV-Hypo, IV-Both.

The OLS coefficient is 0.039 (0.017), significant at the five-percent level. The IV-LP estimate is 0.074 (0.050), the IV-Hypo estimate is 0.192 (0.241), and the combined IV estimate is 0.087 (0.041), significant at the five-percent level. The first-stage F for the IV-Hypo column on this sample of 237 districts is 1.72, below conventional weak-instrument thresholds; the IV-Hypo column is therefore not interpretable in this specification. The first-stage F for IV-LP is 19.83 and for IV-Both is 11.84.

The OLS and IV-Both point estimates for the placebo outcome are non-zero and statistically significant at the five-percent level. Two readings are consistent with the evidence. First, there may be a genuine pre-existing trend correlated with the eventual infrastructure changes of 1960–1986: districts that were growing fastest in 1947–1960 may have been more likely to gain (or lose) market access in the later period. Second, the placebo outcome is defined on a subsample ($N = 237$) that is non-random with respect to the full estimation sample ($N = 311$), and the correlations documented in Section 4.4.1 for the Larkin Plan instrument may interact with that subsample selection.

Section 5.5 reports the main-spec population regression refit on the same 237-district subsample. On that subsample, the IV-Both coefficient on $\Delta \ln \mathrm{MA}_i^{\mathrm{full}}$ is 0.077 (0.036), with $p = 0.032$. This coefficient is similar in magnitude to the placebo coefficient of 0.087 reported above. The two readings offered in the preceding paragraph are therefore not distinguishable by this test: the pre-reform pattern on the 237 subset is of the same order as the post-reform main-spec coefficient on the same subset. We retain the main specification on the full sample as the headline estimate and report the placebo-subsample result in the robustness panel.

4.4.3 First-Stage Results

Table 5 reports the first-stage regressions. Results are described in Section 5.1.

4.5 Additional Robustness Approaches

We assess robustness by varying the distance decay parameter (θ), the set of controls, the sample (excluding major cities, border regions), and the time period (1960–1970, 1970–1991). Results are reported in Table 23.

Table 4: Pre-trends placebo: does $\Delta \ln \mathrm{MA}^{\mathrm{full}}$ predict 1947–1960 population growth?

	(1) OLS	(2) IV-LP	(3) IV-Hypo	(4) IV-Both
$\Delta \ln \mathrm{MA}^{\mathrm{full}}$	0.039**	0.074	0.192	0.087**
	(0.017)	(0.050)	(0.241)	(0.041)
Observations	237	237	237	237
First-stage F	—	19.8	1.7	11.8

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: Dependent variable: $\Delta \ln(\mathrm{pop}_{1960}/\mathrm{pop}_{1947})$. Robust (HC1) SE in parentheses. All columns include baseline log MA (1960), baseline log pop (1960), and the six standardized geographic controls. A near-zero and insignificant coefficient is evidence that post-reform $\Delta \ln \mathrm{MA}$ is not picking up pre-reform trends. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 5: First stage: instrument strength

	(1) LP only	(2) Hypo only
$\Delta \ln \mathrm{MA}^{\mathrm{LP}}$	0.253***	
	(0.053)	
$\Delta \ln \mathrm{MA}^{\mathrm{hypo}}$		0.144**
		(0.070)
Log MA, 1960	-0.235***	-0.224***
	(0.032)	(0.040)
Log population, 1960	-0.037	-0.090
	(0.129)	(0.129)
Elevation (std)	-0.028	0.134
	(0.213)	(0.214)
Ruggedness (std)	0.231	0.313
	(0.220)	(0.227)
Wheat suitability (std)	-0.913***	-1.079***
	(0.228)	(0.245)
Caloric pot. pre-1500 (std)	-2.545***	-3.347***
	(0.952)	(0.984)
Caloric pot. post-1500 (std)	3.513***	4.392***
	(1.096)	(1.150)
Distance to B.A. (std)	0.034	-0.111
	(0.233)	(0.225)
Constant	-9.279***	-8.462***
	(2.291)	(2.604)
Observations	311	311
Adj. R^2	0.332	0.299
First-stage F (instruments)	22.64	4.27

* p < 0.1, ** p < 0.05, *** p < 0.01

Dependent variable: $DeltalnmathrmMA^{mathrmfull}$ (sector $s = 0$, $theta = 4.55$). Robust (HC1) standard errors

5 Main Results

5.1 First Stage

Table 5 reports the first-stage regressions corresponding to equation (3). Column (1) uses only the Larkin Plan instrument $\Delta \ln \mathrm{MA}_i^{\mathrm{LP}}$, column (2) uses only the hypothetical-road instrument $\Delta \ln \mathrm{MA}_i^{\mathrm{hypo}}$, and column (3) uses both jointly. All columns include baseline log market access, baseline log population, and the six standardized geographic controls. Standard errors are heteroskedasticity-robust.

The Larkin Plan instrument has a coefficient of 0.253 with a robust standard error of 0.053 in column (1). The corresponding first-stage F statistic is 22.64. The hypothetical-road instrument has a coefficient of 0.144 (0.070) in column (2), with a first-stage F of 4.27. In the joint specification in column (3), both instruments retain predictive power: the Larkin Plan coefficient is 0.266 (0.054) and the hypothetical-road coefficient is 0.164 (0.063); the joint F is 16.17.

The Larkin Plan instrument alone exceeds the Stock–Yogo rule-of-thumb threshold of 10. The hypothetical-road instrument alone does not. The joint specification exceeds the threshold and is adopted as the main specification throughout this section. The hypothetical-road-only column is reported so that the two sources of variation can be inspected separately.

5.2 Population Effects

Table 6 reports estimates of equation (2) for four population outcomes: the log change in total population, in urban population, and in rural population between 1960 and 1991, and the level change in the urban share over the same period. Each outcome panel reports four columns: OLS in column (1), instrumental variables using the Larkin Plan instrument in column (2), the hypothetical-road instrument in column (3), and both instruments in column (4).

For total population (Panel A), the OLS coefficient on $\Delta \ln \mathrm{MA}_i^{\mathrm{full}}$ is 0.025 (0.012), significant at the five-percent level. The IV-LP estimate is 0.046 (0.040), the IV-Hypo estimate is 0.069 (0.068), and the combined IV estimate is 0.052 (0.031). The sample is 311 districts: the 312 mainland districts less Capital Federal, which is excluded as an observation (Section 3).

For urban population (Panel B), the OLS coefficient is 0.031 (0.013), significant at the five-percent level. The IV-LP estimate is 0.064 (0.045), the IV-Hypo estimate is 0.056 (0.100), and the combined IV estimate is 0.063 (0.038). The sample is 286 districts, reflecting districts with positive urban population in both census years.

For rural population (Panel C), the OLS coefficient is 0.033 (0.025). The IV-LP estimate is 0.110 (0.106), the IV-Hypo estimate is 0.127 (0.170), and the combined IV estimate is 0.116 (0.087). The sample is 272 districts.

For the urban share (Panel D), all four estimates are close to zero and statistically indistinguishable from zero (OLS: 0.001 (0.005); IV-Both: -0.005 (0.016)). A one-unit increase in $\Delta \ln \mathrm{MA}_i^{\mathrm{full}}$ corresponds to a change in the urban share of less than one percentage point.

Across the three log-change outcomes, the IV point estimates exceed the OLS estimates by a

Table 6: Outcome: $\Delta \ln \text{Pop}$

	(1) OLS	(2) IV-LP	(3) IV-Hypo	(4) IV-Both
$\Delta \ln \text{MA}^{\text{full}}$	0.025**	0.046	0.069	0.052*
	(0.012)	(0.040)	(0.068)	(0.031)
Observations	311	311	311	311
First-stage F	—	22.1	6.9	16.2

* p < 0.1, ** p < 0.05, *** p < 0.01

factor of approximately two to four. This pattern is consistent with a downward bias in OLS that would arise if road-driven market-access gains were directed to districts with declining pre-period trends — the compensatory allocation logic of Section 4. The IV point estimates are not statistically distinguishable from zero at the five-percent level, though the total-population combined-IV estimate is significant at the ten-percent level ($p = 0.096$) and the total-population OLS estimate at the five-percent level. The IV-Both column reports the narrowest confidence intervals among the three IV columns.

Two mechanisms can produce an IV–OLS gap of this direction, and the evidence does not separate them. The first is compensatory selection (Section 4): if road-driven market-access gains were directed to districts whose populations would have grown slowly anyway, gains align with negative outcome shocks and OLS understates the causal response. The opposing efficiency logic — closures concentrated where outcomes were already falling — biases OLS upward and so cannot explain a gap of this direction. The placebo evidence bears on which logic dominated: on the subsample where pre-period growth is observed, post-reform market-access changes correlate *positively* with 1947–1960 growth (Section 4.4.2), the efficiency-selection pattern, which taken alone points to upward, not downward, OLS bias and so weighs against selection of either sign explaining the observed gap. The second mechanism is attenuation: market access is a constructed regressor built from digitised networks and calibrated cost parameters, and classical measurement error in it biases OLS toward zero. Instrumenting removes the attenuation only to the extent that the instruments do not share the construction error — a condition that is not innocuous here, since both instruments are themselves market-access measures built from the same digitised baseline network and the same cost parameters. With the available data we do not apportion the gap between the two mechanisms.

Cargo-density cost schedules. The estimates above use the medium-density (500 t/day) cost schedule, under which rail and road costs per ton-km are nearly identical (Table 2), so the market-access index does not build in a preference for either mode. Table 10 re-estimates the total-population specifications under the other two Baumgartner and Palazzo (1969) schedules, switching every market-access object — treatment, both instruments, and the baseline log-MA control — to

Table 7: Outcome: $\Delta \ln \text{Pop}^{\text{urban}}$

	(1) OLS	(2) IV-LP	(3) IV-Hypo	(4) IV-Both
$\Delta \ln \text{MA}^{\text{full}}$	0.031**	0.064	0.056	0.063
	(0.013)	(0.045)	(0.100)	(0.038)
Observations	286	286	286	286
First-stage F	—	23.0	3.6	14.7

* p < 0.1, ** p < 0.05, *** p < 0.01

Table 8: Outcome: $\Delta \ln \text{Pop}^{\text{rural}}$

	(1) OLS	(2) IV-LP	(3) IV-Hypo	(4) IV-Both
$\Delta \ln \text{MA}^{\text{full}}$	0.033	0.110	0.127	0.116
	(0.025)	(0.106)	(0.170)	(0.087)
Observations	272	272	272	272
First-stage F	—	11.6	6.1	9.9

* p < 0.1, ** p < 0.05, *** p < 0.01

Table 9: Outcome: Δ(Urban share)

	(1) OLS	(2) IV-LP	(3) IV-Hypo	(4) IV-Both
$\Delta \ln \text{MA}^{\text{full}}$	0.001	0.003	-0.026	-0.005
	(0.005)	(0.018)	(0.040)	(0.016)
Observations	311	311	311	311
First-stage F	—	22.1	6.9	16.2

* p < 0.1, ** p < 0.05, *** p < 0.01

the row's schedule. The treatment correlates 0.97 (high density) and 0.90 (low density) with its baseline version, so the three rows are related but not interchangeable measures. Reading down the table, the combined-IV estimate falls monotonically as the cost schedule shifts from road-favouring to rail-favouring: 0.087 (0.045, $p = 0.051$) at low density, 0.052 (0.031) at the baseline, and 0.026 (0.024) at high density. The combined-IV first-stage F is 26.05, 16.17, and 19.60 respectively; identification is strongest under the low-density schedule.

Table 10: Population Elasticity Across Cargo-Density Cost Schedules (outcome: $\Delta \ln \mathrm{Pop}_{1960\to1991}$)

Cost schedule	(1) OLS	(2) IV-LP	(3) IV-Hypo	(4) IV-Both
Low density (100 t/day)	0.044**	0.086	0.088*	0.087*
	(0.019)	(0.103)	(0.051)	(0.045)
Medium density (500 t/day, baseline)	0.025**	0.046	0.069	0.052*
	(0.012)	(0.040)	(0.068)	(0.031)
High density (1,000 t/day)	0.012	0.028	0.020	0.026
	(0.010)	(0.030)	(0.057)	(0.024)

Notes: Each row re-estimates the four Table 6 specifications with every market-access object — the treatment, both instruments, and the baseline log-MA control — constructed under the row's Baumgartner and Palazzo (1969) cargo-density cost schedule (Table 2). Geographic controls, baseline log population, and $\theta = 4.55$ are identical across rows. $N = 311$. Robust (HC1) SE. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

5.3 Sectoral Activity

Table 12 reports estimates of equation (2) for five sectoral outcomes drawn from the industrial and agricultural censuses. Panel A covers three manufacturing outcomes from the 1954 and 1985 industrial censuses: the log change in the number of establishments, in the value of production, and in the total wage mass. Panel B covers two agricultural outcomes from the 1960 and 1988 agricultural censuses: the log change in the number of farms and in the total farmed area. Column structure matches Table 6.

For manufacturing establishments, the OLS estimate is 0.029 (0.018) and the combined IV estimate is 0.067 (0.060). Neither is statistically distinguishable from zero.

For manufacturing production value, the OLS estimate is 0.086 (0.040), significant at the five-percent level. The IV-LP estimate is 0.269 (0.136), significant at the five-percent level. The combined IV estimate is 0.317 (0.120), significant at the one-percent level. The sample is 310 districts.

For manufacturing wage mass, the OLS estimate is 0.116 (0.038), significant at the one-percent level. The IV-LP estimate is 0.283 (0.141), significant at the five-percent level. The combined IV estimate is 0.378 (0.124), significant at the one-percent level. The sample is 309 districts.

For the number of agricultural farms (Panel B), the OLS estimate is −0.010 (0.012). The combined IV estimate is 0.096 (0.078). Neither is statistically distinguishable from zero.

For total agricultural farmed area, the OLS estimate is −0.001 (0.018). The combined IV estimate is 0.102 (0.077). Neither is statistically distinguishable from zero.

The estimates differ substantially across the two panels. The manufacturing outcomes related to activity intensity — production value and wage mass — respond strongly and positively to

$\Delta \ln \mathrm{MA}_i^{\mathrm{full}}$: combined-IV point estimates are 0.317 and 0.378, respectively. The number of manufacturing establishments does not respond detectably. The agricultural outcomes show point estimates close to zero with wide confidence intervals. Section 7 returns to the question of whether this pattern is consistent with the scale-economies hypothesis introduced in Section 2.

Sector-matched cost schedules. The estimates above use the medium-density cost schedule for all outcomes. The Baumgartner and Palazzo (1969) schedules assign each sector a cargo profile: bulk agricultural freight travels at high density, where rail is the cheap mode, and small-batch manufacturing freight at low density, where road is (Table 2). Table 11 re-estimates each sectoral outcome with the market-access measure constructed under its matched schedule, switching the treatment, both instruments, and the baseline log-MA control together. The manufacturing estimates strengthen under their matched (road-favouring) measure: production value rises from 0.317 to 0.367 (0.157) and wage mass from 0.378 to 0.444 (0.183), with combined-IV first-stage F of 26.4 and 25.8 respectively; the number of establishments remains statistically indistinguishable from zero, as at the baseline schedule. The agricultural estimates remain statistically indistinguishable from zero under their matched (rail-favouring) measure (farms 0.059, farmed area 0.060; $F = 18.6$). The sectoral contrast of the previous paragraph is therefore not an artifact of evaluating both sectors at the general-purpose cost schedule.

Table 11: Sectoral Outcomes Under Sector-Matched Cost Schedules

Outcome	(1) OLS	(2) IV-LP	(3) IV-Hypo	(4) IV-Both	F (IV-B)	N
Panel A: manufacturing outcomes, low-density (100 t/day) MA						
Mfg. establishments	0.052*	0.080	0.131	0.123	26.4	310
	(0.029)	(0.223)	(0.091)	(0.084)		
Mfg. production value	0.121**	0.769*	0.296*	0.367**	26.4	310
	(0.060)	(0.409)	(0.168)	(0.157)		
Mfg. wage mass	0.148**	0.680	0.408**	0.444**	25.8	309
	(0.064)	(0.492)	(0.196)	(0.183)		
Panel B: agricultural outcomes, high-density (1,000 t/day) MA						
Ag. farms	-0.009	0.063	0.038	0.059	18.6	297
	(0.009)	(0.066)	(0.078)	(0.054)		
Ag. farmed area	-0.002	0.046	0.128	0.060	18.6	297
	(0.013)	(0.061)	(0.123)	(0.052)		

Notes: Each row re-estimates the corresponding Table 12 specification with every market-access object — treatment, both instruments, and the baseline log-MA control — constructed under the cost schedule matched to the sector's cargo profile (Table 2): low density for manufacturing (small-batch goods), high density for agriculture (bulk grain). Geographic controls, baseline log population, and $\theta = 4.55$ are identical to Table 12, and sample sizes match it row-for-row (asserted in the generating script). F is the combined-IV first-stage Wald F. Robust (HC1) SE. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Three facts connect this sectoral pattern to the cost structure described in Section 2.4. First, the Baumgartner and Palazzo (1969) schedule makes the modal shift asymmetric across sectors by construction: moving freight from rail to road lowers unit costs at low cargo density (8.266 versus 13.490 1960 pesos per ton-km) and raises them at high density (1.000 versus 0.465), so a restructuring that

Table 12: A. Manufacturing (industrial census 1954-1985). Outcome: Mfg. establishments

	(1) OLS	(2) IV-LP	(3) IV-Hypo	(4) IV-Both
$\Delta \ln \mathrm{MA}^{\mathrm{full}}$	0.029	0.059	0.088	0.067
	(0.018)	(0.073)	(0.120)	(0.060)
Observations	310	310	310	310
First-stage F	—	22.5	7.0	16.5

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 13: A. Manufacturing (industrial census 1954-1985). Outcome: Mfg. production value

	(1) OLS	(2) IV-LP	(3) IV-Hypo	(4) IV-Both
$\Delta \ln \mathrm{MA}^{\mathrm{full}}$	0.086**	0.269**	0.451	0.317***
	(0.040)	(0.136)	(0.294)	(0.120)
Observations	310	310	310	310
First-stage F	—	22.5	7.0	16.5

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

replaced rail with road capacity changed manufacturing's transport costs in the opposite direction from agriculture's. Second, the outcomes align with that asymmetry: the sector whose characteristic shipments became cheaper to move responds strongly (Table 12), and the response strengthens when its market access is measured under its own cost schedule (Table 11), while the sector whose shipments became more expensive shows no response under either measurement. Third, the same gradient appears for the aggregate outcome: the population elasticity rises monotonically as the cost schedule shifts from rail-favouring to road-favouring (Table 10). The interpretive weighing of these facts — what they imply about scale economies and the modal shift as industrial policy — is deferred to Section 8.3.

The manufacturing IV–OLS gaps run in the same direction as the population gap of Section 5.2 and are larger in proportion; the two mechanisms discussed there — selection of closures into declining districts and measurement error in the constructed regressor — apply to the sectoral outcomes as well, and we do not apportion between them here either.

5.4 Other Outcomes

Table 17 reports estimates of equation (2) for four outcomes drawn from IPUMS microdata: the level change in the share of adults with college education, the level change in the share with secondary education, the level change in the share of residents who migrated in the past five years, and the level change in the employment rate (the share of individuals classified as employed in IPUMS

Table 14: A. Manufacturing (industrial census 1954-1985). Outcome: Mfg. wage mass

	(1) OLS	(2) IV-LP	(3) IV-Hypo	(4) IV-Both
$\Delta \ln \mathrm{MA}^{\mathrm{full}}$	0.116***	0.283**	0.657*	0.378***
	(0.038)	(0.141)	(0.343)	(0.124)
Observations	309	309	309	309
First-stage F	—	23.6	7.0	17.1

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 15: B. Agriculture (agricultural census 1960-1988). Outcome: Ag. farms

	(1) OLS	(2) IV-LP	(3) IV-Hypo	(4) IV-Both
$\Delta \ln \mathrm{MA}^{\mathrm{full}}$	-0.010	0.104	0.066	0.096
	(0.012)	(0.101)	(0.101)	(0.078)
Observations	297	297	297	297
First-stage F	—	22.2	5.0	15.3

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 16: B. Agriculture (agricultural census 1960-1988). Outcome: Ag. farmed area

	(1) OLS	(2) IV-LP	(3) IV-Hypo	(4) IV-Both
$\Delta \ln \mathrm{MA}^{\mathrm{full}}$	-0.001	0.095	0.130	0.102
	(0.018)	(0.095)	(0.135)	(0.077)
Observations	297	297	297	297
First-stage F	—	22.2	5.0	15.3

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

EMPSTAT). All four outcomes are level differences in population shares between 1970 and 1991. The regressor remains $\Delta \ln \mathrm{MA}_i^{\mathrm{full}}$ computed over 1960–1986; infrastructure restructuring predates the 1970 outcome measurement. Column structure matches Table 6.

One feature of these four outcomes organizes how we read them. For three of the four, the two instruments do not identify a common parameter. The identification-robust overidentification test defined in Section 4 rejects at the 5-percent level for the college share ($p = 0.031$), the secondary share ($p = 0.007$), and the recent-migration share ($p = 0.012$), and does not reject for the employment rate ($p = 0.724$). The classical Sargan statistic reported in Table 17 gives the same four-way split ($p = 0.034$, 0.008, 0.006, and 0.737 respectively), so heteroskedasticity is not what drives the result. We therefore report the combined-instrument estimate as the result for the employment rate, and for the other three outcomes we report what each instrument implies separately.

For the employment rate, the OLS estimate is -0.001 (0.002) and the combined IV estimate is -0.011 (0.005), significant at the five-percent level ($p = 0.032$). The Larkin Plan instrument alone gives -0.012 (0.006, $p = 0.042$). The Anderson–Rubin confidence set for the combined specification, which remains valid whatever the strength of the instruments, is $[-0.0280, -0.0003]$ and excludes zero. The employment rate is measured as a share of individuals, so a one-unit increase in $\Delta \ln \mathrm{MA}_i^{\mathrm{full}}$ is associated with a fall in the employment rate of 1.1 percentage points.

For the college share, every estimate is individually indistinguishable from zero, with point estimates between -0.002 and 0.001; the two instruments nonetheless disagree with each other (0.0010 under the Larkin Plan instrument against -0.0021 under the hypothetical-network instrument), which is what the overidentification test detects. For the secondary share the disagreement is sharper. The Larkin Plan instrument gives 0.006 ($p = 0.007$), with an Anderson–Rubin set of $[0.0019, 0.0114]$ lying entirely above zero, while the hypothetical-network instrument gives -0.010; the OLS and combined-IV point estimates are 0.000 and 0.002. We do not read a schooling effect from these estimates, because the two instruments imply different ones. This is the one cell in the table where the evidence cuts against that reading: taken on its own, the Larkin Plan instrument implies a positive effect on the secondary share.

For the recent-migration share, the two instruments imply estimates of different magnitudes: -0.007 (0.010, $p = 0.518$) under the Larkin Plan instrument, whose Anderson–Rubin set $[-0.0303, 0.0138]$ contains zero, against -0.064 ($p = 0.029$) under the hypothetical-network instrument, which Section 4.4.3 shows to be the weaker of the two. The Anderson–Rubin set for that second estimate, $[-1.0796, -0.0297]$, excludes zero but spans more than an order of magnitude, so it restricts the sign and not the size. We therefore do not treat the negative combined estimate as a finding. Appendix A reports the full set of estimates and the readings that have been proposed for the sign.

The window for these four outcomes is 1970–1991 because Argentine IPUMS microdata begins in 1970, one decade after the start of the estimation window for Table 6. The infrastructure restructuring and the bulk of rail closures occurred before 1970, so the timing of outcome measurement post-dates treatment.

Table 17: Outcome: Δ(Employment rate)

	(1) OLS	(2) IV-LP	(3) IV-Hypo	(4) IV-Both
$\Delta \ln \mathrm{MA}^{\mathrm{full}}$	-0.001	-0.012**	-0.008	-0.011**
	(0.002)	(0.006)	(0.011)	(0.005)
Observations	311	311	311	311
First-stage F	—	22.1	6.9	16.2
Overid. p (Sargan)	—	—	—	0.737

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: Outcomes are level changes in population shares between 1970 and 1991; the regressor is the 1960–1986 change in log market access. All columns include baseline log market access (1960), baseline log population (1960), and the six standardized geographic controls. Robust (HC1) standard errors. "Overid. p" is the classical (homoskedastic) Sargan p-value for the two-instrument column; the identification-robust counterpart and the corresponding Anderson–Rubin confidence sets are reported in Section 5.4.

Table 18: Outcome: Δ(College share)

	(1) OLS	(2) IV-LP	(3) IV-Hypo	(4) IV-Both
$\Delta \ln \mathrm{MA}^{\mathrm{full}}$	0.000	0.001	-0.002	0.000
	(0.000)	(0.001)	(0.002)	(0.001)
Observations	311	311	311	311
First-stage F	—	22.1	6.9	16.2
Overid. p (Sargan)	—	—	—	0.034

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: Sample, controls, standard errors, and the definition of the overidentification row are as in Table 17.

Table 19: Outcome: Δ(Secondary share)

	(1) OLS	(2) IV-LP	(3) IV-Hypo	(4) IV-Both
$\Delta \ln \mathrm{MA}^{\mathrm{full}}$	0.000	0.006***	-0.010	0.002
	(0.001)	(0.002)	(0.007)	(0.002)
Observations	311	311	311	311
First-stage F	—	22.1	6.9	16.2
Overid. p (Sargan)	—	—	—	0.008

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: Sample, controls, standard errors, and the definition of the overidentification row are as in Table 17.

Table 20: Outcome: Δ(Recent-migration share)

	(1) OLS	(2) IV-LP	(3) IV-Hypo	(4) IV-Both
$\Delta \ln \mathrm{MA}^{\mathrm{full}}$	-0.008***	-0.007	-0.064**	-0.022***
	(0.002)	(0.010)	(0.029)	(0.008)
Observations	311	311	311	311
First-stage F	—	22.1	6.9	16.2
Overid. p (Sargan)	—	—	—	0.006

* p < 0.1, ** p < 0.05, *** p < 0.01

Notes: Sample, controls, standard errors, and the definition of the overidentification row are as in Table 17.

5.5 Robustness

Table 23 reports three stress tests of the main population elasticity estimated in Section 5.2 (Panel A of Table 6): the coefficient on $\Delta \ln \mathrm{MA}_i^{\mathrm{full}}$ is 0.052 (0.031) under the combined IV specification on the full 311-district sample.

Panel A — alternative trade elasticity. The main specification sets the trade elasticity $\theta = 4.55$. Panel A reports the same four specifications with $\theta = 8.11$, which corresponds to a higher-elasticity calibration in the market-access literature. All MA variables, including the treatment, the two instruments, and the baseline log-MA control, switch to the $\theta = 8.11$ variant. The combined IV coefficient is 0.024 (0.016). The coefficient is approximately half the main-specification value. The direction of the effect is unchanged, and no coefficient flips sign.

Table 21 extends the two-value comparison to the full sweep $\theta \in [1, 12]$, recomputing market access, both instruments, and the baseline control at each value. The combined-IV coefficient declines monotonically in θ, from 0.36 at $\theta = 1$ to 0.01 at $\theta = 12$, while the first-stage F stays between 12.5 and 16.9 across the grid. The level of the population elasticity is therefore calibration-dependent; its sign is not. The sectoral contrast of Section 5.3 is not calibration-dependent: Table 22 repeats the sweep for the five sectoral outcomes. Manufacturing production value and wage mass are significant at every value of θ in the grid (largest $p = 0.014$), while manufacturing establishments and both agricultural outcomes are null throughout. The pattern the paper emphasizes — manufacturing responds on the intensive margin, agriculture does not — holds at every calibration we examined. Section 8 discusses the implications for comparisons with estimates in the literature.

Transshipment bound. The cost surface underlying all estimates allows free mode switching: a least-cost path can move between rail, road, and navigation at any grid cell at no cost (Section 3.3). The opposite extreme is infinite transshipment: each origin–destination trip uses a single mode plus off-network land, with the trip cost taken as the minimum across the three single-mode cost matrices. Under this bound, the share of districts gaining market access between 1960 and 1986 falls from 90.7% to 77.2%, and the median origin–destination cost ratio (1986 relative to 1960) moves from 0.80

Table 21: Population Elasticity Across the Trade-Elasticity Sweep (outcome: $\Delta \ln \mathrm{Pop}_{1960 \to 1991}$)

θ	OLS	IV-Both	First-stage F
1.00	0.138 (0.074)	0.357 (0.196)	14.6
2.00	0.067 (0.034)	0.156 (0.085)	16.1
3.00	0.043 (0.021)	0.090 (0.051)	16.9
4.55 (main)	0.025 (0.012)	0.052 (0.031)	16.2
6.00	0.016 (0.009)	0.036 (0.023)	14.8
8.11 (alt.)	0.010 (0.006)	0.024 (0.016)	13.4
10.00	0.007 (0.005)	0.019 (0.013)	12.8
12.00	0.006 (0.004)	0.015 (0.011)	12.5

Notes: Each row recomputes market access from the existing transport-cost matrices with the row's θ (treatment, both instruments, and the baseline log-MA control all switch), then re-estimates the OLS and combined-IV specifications of Table 6. $N = 311$. Robust (HC1) SE in parentheses.

Table 22: Sectoral Elasticities Across the Trade-Elasticity Sweep (combined-IV estimates)

	Manufacturing			Agriculture	
θ	Value of prod.	Wage mass	Establishments	Farms	Farmed area
1.00	2.467^{***}	3.033^{***}	0.510	0.771	0.717
	(0.777)	(0.830)	(0.400)	(0.525)	(0.549)
2.00	0.965^{***}	1.156^{***}	0.197	0.302	0.293
	(0.335)	(0.345)	(0.170)	(0.227)	(0.225)
3.00	0.550^{***}	0.652^{***}	0.112	0.169	0.172
	(0.202)	(0.207)	(0.101)	(0.134)	(0.131)
4.55 (main)	0.317^{***}	0.378^{***}	0.067	0.096	0.102
	(0.120)	(0.124)	(0.060)	(0.078)	(0.077)
6.00	0.223^{**}	0.268^{***}	0.048	0.065	0.072
	(0.087)	(0.090)	(0.042)	(0.056)	(0.055)
8.11 (alt.)	0.157^{**}	0.189^{***}	0.034	0.044	0.049
	(0.063)	(0.065)	(0.030)	(0.039)	(0.039)
10.00	0.124^{**}	0.151^{***}	0.027	0.034	0.038
	(0.050)	(0.052)	(0.023)	(0.031)	(0.031)
12.00	0.101^{**}	0.124^{***}	0.022	0.027	0.031
	(0.041)	(0.043)	(0.019)	(0.025)	(0.025)

Notes: Each row recomputes market access from the existing transport-cost matrices with the row's θ (treatment, both instruments, and the baseline log-MA control all switch), then re-estimates the combined-IV specification of Table 12 for each sectoral outcome. $N = 297$–310 by outcome. First-stage F between 11.5 and 17.9 across the grid. Robust (HC1) SE in parentheses. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The population sweep is in Table 21.

to 0.61: the bound reshapes the market-access distribution, but the broad cost decline persists when no trip may switch modes, and the median pair still becomes substantially cheaper. The population elasticity is essentially unchanged at the bound: an OLS regression of the population change on the unimodal market-access change yields 0.018 (0.010), against 0.016 (0.011) for the baseline measure on the same specification and sample ($N = 311$).[9] The truth — finite transshipment costs at stations and ports — lies between the two extremes, which agree on the population elasticity; on this evidence the free-switching assumption is not driving that estimate. The interior case is the object of structural work: Fuchs and Wong (2026) estimate terminal congestion directly and calibrate mode-switching costs at intermodal terminals for the contemporary United States, placing the multimodal freight system strictly between the two extremes bracketed here.

Panel B — alternative hypothetical-road instruments. The main specification uses the least-cost-path minimum spanning tree as the hypothetical-road instrument. Panel B reports results using three alternative hypothetical networks: the Euclidean minimum spanning tree, the full bilateral least-cost path network, and the full bilateral Euclidean network. Holding θ at 4.55, the combined-IV coefficient on $\Delta \ln \mathrm{MA}_i^{\mathrm{full}}$ ranges from 0.027 to 0.070 across the four choices. Individual IV-Hypo columns show wide variation; the large point estimates and standard errors in that column reflect weak-instrument noise in some of the alternative networks and are not interpretable in isolation. The combined-IV column is more stable.

Panel C — sample robustness. Panel C refits the main specification on the 237-district subsample for which the 1947 placebo outcome is defined, as reported in Section 4.4.2. On this subsample, the OLS coefficient is 0.032 (0.013), significant at the five-percent level, and the combined IV coefficient is 0.077 (0.036), significant at the five-percent level. Both estimates are larger than their full-sample counterparts.

The Panel C estimate bears on the interpretation of Section 4.4.2. The placebo coefficient on the 1947–1960 population change was 0.087 on the same 237-district subsample. The main-specification coefficient on the 1960–1991 population change on the same subsample is 0.077. The pre-reform coefficient and the post-reform coefficient are of the same order of magnitude on the common sample. This co-movement is consistent with both candidate readings offered in Section 4.4.2 and does not adjudicate between them.

6 Mode-Specific Patterns via Counterfactuals

The estimates in Section 5 measure the effect of the combined change in market access produced by the simultaneous contraction of the rail network and expansion of the road network. This section decomposes that combined change into its two modal components. The decomposition holds one

[9] The bound is evaluated by OLS only: instrumenting would require unimodal versions of both instruments, and the comparison specification omits the baseline log-MA control on both sides (no unimodal analogue exists), so the baseline OLS coefficient here differs from Table 6 column (1).

Table 23: Robustness of the main population elasticity (outcome: $\Delta \ln \text{Pop}_{1960 \to 1991}$)

	Variation	(1) OLS	(2) IV-LP	(3) IV-Hypo	(4) IV-Both
Panel A: alternative trade elasticity					
A	Alt. theta = 8.11	0.010* (0.006)	0.021 (0.022)	0.030 (0.032)	0.024 (0.016)
Panel B: alternative hypothetical-road instruments					
B	Hypo = LCP-MST (main)	0.025** (0.012)	0.046 (0.040)	0.069 (0.068)	0.052* (0.031)
B	Hypo = Euclidean-MST	0.025** (0.012)	0.046 (0.040)	0.451 (0.458)	0.070* (0.040)
B	Hypo = LCP (bilateral)	0.025** (0.012)	0.046 (0.040)	-0.076 (0.119)	0.027 (0.033)
B	Hypo = Euclidean (bilateral)	0.025** (0.012)	0.046 (0.040)	0.036 (0.076)	0.043 (0.036)
Panel C: sample robustness					
C	Placebo subsample (N=237)	0.032** (0.013)	0.063 (0.041)	0.195 (0.182)	0.077** (0.036)
C	Full sample (for reference)	0.025** (0.012)	0.046 (0.040)	0.069 (0.068)	0.052* (0.031)

Notes: All regressions use $\Delta \ln(\text{Pop}_{1991}/\text{Pop}_{1960})$ as the outcome. Panel A swaps the MA elasticity from $\theta = 4.55$ to $\theta = 8.11$ (all MA variables, treatment, instruments, and baseline control, switch to `_ehigh` variants). Panel B holds θ at 4.55 and replaces the LCP-MST hypothetical-road instrument with one of three alternatives. Panel C refits the main specification on the 237-district subset for which the 1947 placebo outcome is defined. Robust (HC1) SE. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

network fixed at its pre-reform configuration while the other changes as observed, and recomputes market access under each hypothetical configuration.

6.1 Counterfactual Market Access Measures

We construct two counterfactual market access measures using the same cost-surface pipeline as the actual measures described in Section 3. The *only-rail* measure, $\Delta \ln \text{MA}_i^{\text{only-rail}}$, combines the 1986 rail network with the road network frozen at its 1954 configuration; it isolates the market-access consequences of the rail contraction. The *only-road* measure, $\Delta \ln \text{MA}_i^{\text{only-road}}$, combines the road network of 1986 with the rail network frozen at its 1960 configuration; it isolates the consequences of the road expansion. Both measures use the same trade elasticity ($\theta = 4.55$), the same destination populations, and the same navigation layer as the full measure, so differences across the three measures reflect only the network configuration.

Each counterfactual regressor is instrumented with the instrument that targets its source of variation. The only-rail measure is instrumented with the Larkin Plan instrument alone: the Plan's study designations shift rail-driven market access and have no mechanical connection to the frozen road layer. The only-road measure is instrumented with the hypothetical-road instrument alone. The full measure, reproduced from Table 6 for comparison, uses both instruments. The three regressors are never entered jointly in one regression, for two reasons. The full change is by construction close to the sum of its two components, so the three measures are nearly collinear. And regressions with multiple treatments generally fail to recover convex averages of heterogeneous effects: each coefficient is contaminated by a non-convex average of the other treatments' effects (Goldsmith-Pinkham et al., 2024). Estimating each measure in a separate regression avoids both problems.

Figure 3 maps the three measures on a common scale. The only-rail change is negative in every district: the median district loses three log points of market access from the rail contraction alone (interquartile range seven to two log points), and a small number of remote districts lose substantially more. The only-road change is positive for nearly all districts and reproduces the broad spatial pattern of the full measure, consistent with the road expansion being the dominant source of the aggregate market-access gains documented in Section 3.

6.2 Results

Table 24 reports estimates of the analogue of equation (2) with each of the three regressors, for the four population outcomes of Table 6. Panel A reproduces the full-measure estimates. Panels B and C report the only-rail and only-road estimates.

For total population, the IV estimate on the only-rail measure is 0.036 (0.034) with a first-stage F of 108.6. The IV estimate on the only-road measure is 0.050 (0.047) with a first-stage F of 15.6. The full-measure estimate is 0.052 (0.031) with an F of 16.2. The three point estimates are of similar magnitude; none is statistically distinguishable from zero at conventional levels, and none is distinguishable from the others. What differs sharply across panels is identification strength: the Larkin Plan instrument predicts rail-driven market access far more strongly (F between 70.6 and 122.3 across outcomes) than either instrument predicts its respective measure in the other panels.

The urban and rural decomposition shows the same pattern. Rail-only estimates are 0.048 (0.035) for urban and 0.080 (0.083) for rural population; only-road estimates are 0.035 (0.060) and 0.104 (0.136). The urban-share estimates are near zero in all three panels, as in Table 6. OLS estimates differ across panels: the only-rail OLS coefficient is -0.003 (0.019), while IV moves it to 0.036, a pattern consistent with rail lines having been closed in districts that were already declining.

Table 25 repeats the decomposition for the five sectoral outcomes of Table 12. Manufacturing responds through both modal channels: the only-rail IV estimates are 0.215 (0.125) for value of production and 0.222 (0.125) for wage mass, and the only-road estimates are 0.323 (0.175) and 0.471 (0.188). For each outcome the only-rail and only-road estimates are not statistically distinguishable from each other. Manufacturing establishments and both agricultural outcomes show no detectable response in any panel, matching the intensive-margin pattern of Table 12.

6.3 Interpretation and Caveats

Three features of this exercise limit what the estimates can support, and we state them before drawing any conclusion from the comparison.

First, the counterfactual networks are accounting constructs, not equilibrium predictions. Freezing the road network at 1954 does not model how firms, households, or the government would have

Table 24: Counterfactual decomposition of the population effect

	(1) OLS	(2) IV
A. Full MA: $\Delta \ln \mathrm{MA}^{\mathrm{full}}$ *(both modes change)*		
Instrument in column (2): IV-Both (LP and Hypo)		
$\Delta \ln \mathrm{Pop}$	0.025^{**}	0.052^{*}
	(0.012)	(0.031)
$\Delta \ln \mathrm{Pop}^{\mathrm{urban}}$	0.031^{**}	0.063
	(0.013)	(0.038)
$\Delta \ln \mathrm{Pop}^{\mathrm{rural}}$	0.033	0.116
	(0.025)	(0.087)
Δ(Urban share)	0.001	-0.005
	(0.005)	(0.016)
First-stage F (by outcome)	—	16.2 / 14.7 / 9.9 / 16.2
Observations (by outcome)	—	311 / 286 / 272 / 311
B. Only-rail MA: $\Delta \ln \mathrm{MA}^{\mathrm{only\_rail}}$ *(roads frozen at 1954)*		
Instrument in column (2): IV-LP (Larkin Plan)		
$\Delta \ln \mathrm{Pop}$	-0.003	0.036
	(0.019)	(0.034)
$\Delta \ln \mathrm{Pop}^{\mathrm{urban}}$	-0.002	0.048
	(0.019)	(0.035)
$\Delta \ln \mathrm{Pop}^{\mathrm{rural}}$	0.041	0.080
	(0.033)	(0.083)
Δ(Urban share)	0.001	0.002
	(0.005)	(0.014)
First-stage F (by outcome)	—	108.6 / 122.3 / 70.6 / 108.6
Observations (by outcome)	—	311 / 286 / 272 / 311
C. Only-road MA: $\Delta \ln \mathrm{MA}^{\mathrm{only\_road}}$ *(rails frozen at 1960)*		
Instrument in column (2): IV-Hypo (LCP-MST)		
$\Delta \ln \mathrm{Pop}$	0.035^{***}	0.050
	(0.012)	(0.047)
$\Delta \ln \mathrm{Pop}^{\mathrm{urban}}$	0.041^{***}	0.035
	(0.013)	(0.060)
$\Delta \ln \mathrm{Pop}^{\mathrm{rural}}$	0.030	0.104
	(0.027)	(0.136)
Δ(Urban share)	0.001	-0.019
	(0.006)	(0.027)
First-stage F (by outcome)	—	15.6 / 11.1 / 9.9 / 15.6
Observations (by outcome)	—	311 / 286 / 272 / 311

Notes: Each cell is one regression of the row outcome on the panel's $\Delta \ln \mathrm{MA}$ measure. Column (1) is OLS; column (2) instruments the MA measure with the panel-specific instrument noted in italics. Panel A reports the joint IV-Both (replicating Table 6). Panel B uses the Larkin Plan instrument; Panel C uses the LCP-MST hypothetical-road instrument. Each panel's first-stage F row lists F for the four outcomes in the same order as the outcome rows. All regressions include baseline log MA (1960), baseline log pop (1960), and the six standardized geographic controls. Robust (HC1) standard errors. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 25: Counterfactual decomposition of the sectoral effects

	(1) OLS	(2) IV
A. Full MA: $\Delta \ln \text{MA}^{\text{full}}$ *(both modes change)*		
Instrument in column (2): IV-Both (LP and Hypo)		
Δ ln(Value of production)	0.086**	0.317***
	(0.040)	(0.120)
Δ ln(Wage mass)	0.116***	0.378***
	(0.038)	(0.124)
Δ ln(Establishments)	0.029	0.067
	(0.018)	(0.060)
Δ ln(Farms)	-0.010	0.096
	(0.012)	(0.078)
Δ ln(Farmed area)	-0.001	0.102
	(0.018)	(0.077)
First-stage F (by outcome)	—	16.5 / 17.1 / 16.5 / 15.3 / 15.3
Observations (by outcome)	—	310 / 309 / 310 / 297 / 297
B. Only-rail MA: $\Delta \ln \text{MA}^{\text{only\_rail}}$ *(roads frozen at 1954)*		
Instrument in column (2): IV-LP (Larkin Plan)		
Δ ln(Value of production)	0.073	0.215*
	(0.059)	(0.125)
Δ ln(Wage mass)	0.069	0.222*
	(0.044)	(0.125)
Δ ln(Establishments)	0.003	0.047
	(0.026)	(0.059)
Δ ln(Farms)	-0.029**	0.077
	(0.014)	(0.078)
Δ ln(Farmed area)	-0.019	0.069
	(0.013)	(0.073)
First-stage F (by outcome)	—	108.4 / 121.6 / 108.4 / 125.7 / 125.7
Observations (by outcome)	—	310 / 309 / 310 / 297 / 297
C. Only-road MA: $\Delta \ln \text{MA}^{\text{only\_road}}$ *(rails frozen at 1960)*		
Instrument in column (2): IV-Hypo (LCP-MST)		
Δ ln(Value of production)	0.100**	0.323*
	(0.041)	(0.175)
Δ ln(Wage mass)	0.128***	0.471**
	(0.040)	(0.188)
Δ ln(Establishments)	0.039**	0.063
	(0.018)	(0.083)
Δ ln(Farms)	-0.001	0.044
	(0.013)	(0.063)
Δ ln(Farmed area)	0.006	0.087
	(0.019)	(0.083)
First-stage F (by outcome)	—	15.9 / 15.9 / 15.9 / 12.9 / 12.9
Observations (by outcome)	—	310 / 309 / 310 / 297 / 297

Notes: Each cell is one regression of the row outcome on the panel's $\Delta \ln \text{MA}$ measure, mirroring Table 24 for the sectoral outcomes of Table 12. Column (1) is OLS; column (2) instruments the MA measure with the panel-specific instrument noted in italics. Panel A reproduces the IV-Both estimates of Table 12. The two counterfactual measures are never in the same regression. Each panel's first-stage F and observation rows list the five outcomes in row order. All regressions include baseline log MA (1960), baseline log pop (1960), and the six standardized geographic controls. Robust (HC1) standard errors. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

behaved had the road program not occurred. The estimates therefore characterize the relative importance of the two modal shocks for the market-access channel; they are not estimates of the effect of a counterfactual policy.

Second, the exclusion restrictions are inherited from Section 4 and are not separately testable here. The Larkin Plan instrument is excludable for the only-rail measure under the same conditions as for the full measure. For the only-road measure, the hypothetical-network instrument's first-stage F of 15.6 for total population falls to 11.1 for urban population and ranges from 12.9 to 15.9 across the sectoral outcomes, so the only-road panel is less precisely identified throughout.

Third, differences in precision across panels reflect the instruments, not the economics. The only-rail panel is the best-identified object in this paper ($F = 108.6$ for total population, 108.4–125.7 across the sectoral outcomes); this makes its estimates the most credible, not the largest or the most important.

With these limits stated, the decomposition supports one suggestive conclusion: both modal shocks contribute to the outcomes that respond at all. Manufacturing is the clearest case: value of production and wage mass are the only outcomes with estimates significant in both counterfactual panels. For population the point estimates are similar across panels, and the rail component is identified with far greater precision. The estimates give no support to the view that one mode's market-access channel dominates the other; the data are instead consistent with activity responding to market access itself, whichever mode produces it. In the vocabulary of Fuchs and Wong (2026), the episode combines modal substitution — traffic shifting from rail to road where both served the same connections — with modal complementarity, as the cheaper road mode lowered composite transport costs and raised market access broadly; our decomposition measures the combined market-access consequence of each modal shock rather than separating the two margins. Their structural counterfactual of removing the contemporary United States rail network in full is a modern analogue, at greater severity, of the partial contraction Argentina actually carried out. Section 7 examines how much of this response operates through infrastructure located in the district itself.

7 Mechanisms and Heterogeneity

Market access summarizes a district's transport position relative to every other district. A change in a district's market access can therefore reflect infrastructure built or removed within the district itself, or changes elsewhere in the network that alter the district's effective distance to population. This section separates the two, and then asks whether the population response varies with baseline district characteristics.

7.1 Local Infrastructure

We measure within-district infrastructure change with four variables, denoted Z_i: the change in rail kilometers between 1960 and 1986; the change in paved and gravel road kilometers between 1954 and 1986; an indicator for districts that lost their entire rail network by 1986 (14 districts); and an indicator for districts that gained their first paved or gravel road (82 districts). The kilometer measures are computed from the same georeferenced networks as the market access measures.

Table 26 adds the Z_i progressively to the headline specification of Table 6, column (4). The market-access regressor remains instrumented by both instruments; the Z_i enter as controls and are not instrumented. Column (1) reproduces the baseline estimate of 0.052 (0.031). Adding the two kilometer measures in column (2) lowers the market-access coefficient to 0.035 (0.030); adding the two binary margins alone in column (3) leaves it at 0.058 (0.030); and including all four Z_i jointly in column (4) lowers it to 0.029 (0.030). The first-stage F is higher in every augmented column (up to 20.2) than at baseline (16.2), so the attenuation is not a weak-instrument artifact. Columns (2)–(4) lose three districts with missing kilometer measures (308 versus 311 observations); re-estimated on the common 308-district sample, the baseline coefficient is 0.062 (0.031), so the sample change does not generate the attenuation and, if anything, works against it.

Table 26: Local-infrastructure mechanism: progressive-add of Z_i to the headline IV-Both regression

	(1) Baseline	(2) + Δkm rail/road	(3) + binary margins	(4) all Z_i
$\Delta \ln \mathrm{MA}^{\mathrm{full}}$	0.052* (0.031)	0.035 (0.030)	0.058* (0.030)	0.029 (0.030)
Local infrastructure Z_i:				
Δ rail km (1960-1986)		0.0013 (0.0004)		0.0015 (0.0004)
Δ road km (1954-1986)		-0.0001 (0.0001)		-0.0001 (0.0001)
Lost all rails by 1986			0.0925 (0.1160)	0.1952 (0.1130)
Gained first paved or gravel road			-0.0518 (0.0642)	-0.0369 (0.0613)
First-stage F	16.2	18.6	17.9	20.2
Observations	311	308	308	308

Notes: Each column is one IV regression of $\Delta \ln(\mathrm{Pop}_{1991}/\mathrm{Pop}_{1960})$ on $\Delta \ln \mathrm{MA}_i^{\mathrm{full}}$ (instrumented by both the Larkin Plan and the LCP-MST hypothetical-road instruments) plus the local infrastructure controls listed. Local controls are not instrumented. Standard controls (baseline log MA, baseline log pop, and the six standardized geographic controls) are partialled out and not displayed. Robust (HC1) standard errors. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Read as a decomposition, column (4) indicates that roughly half of the headline market-access coefficient operates through infrastructure located in the district itself: conditioning on own rail and road change and the two binary margins removes about half of the point estimate. The change in own rail kilometers enters positively and precisely, at 0.0015 (0.0004) in column (4): districts that lost more rail kilometers grew more slowly, conditional on market access. The change in own road kilometers enters with a coefficient indistinguishable from zero.

The same numbers support a complementary reading. Even conditional on every measured form of own-district infrastructure change, half the market-access coefficient remains. Market access is therefore not merely a repackaging of local infrastructure counts; it carries information about regional connectivity beyond the district's own kilometers. The two readings describe the same estimate and are not alternatives between which the data can select.

Three caveats apply. First, because the Z_i are not instrumented, column (4) is a descriptive decomposition, not a causal mediation analysis: own-infrastructure change is itself an outcome of the same reform. Second, when several treatment-like regressors enter one regression, each coefficient can absorb a non-convex average of the others' heterogeneous effects (Goldsmith-Pinkham et al., 2024); this reinforces reading the column as description rather than as a set of separately identified effects. Third, the indicator for losing the entire rail network enters positively, at 0.195 (0.113), in column (4). With only 14 districts in that group, we do not interpret this coefficient beyond noting it as a candidate for future work with richer within-district data.

7.2 Heterogeneity

If market access matters more where initial conditions leave more room for reallocation, its coefficient should vary with baseline characteristics. We examine three, one at a time, each standardized within the estimation sample: initial log population in 1960, the rural population share in 1960, and distance to Buenos Aires. Under the reallocation hypothesis the interaction should be negative for initial population and positive for the rural share and for distance to Buenos Aires. Each specification adds the interaction of the characteristic with $\Delta \ln \mathrm{MA}_i^{\mathrm{full}}$ to the main equation, instrumenting the interaction with the interactions of both instruments with the characteristic.[10]

The OLS interaction coefficients carry the predicted sign in all three cases: -0.025 (0.012) for initial population, 0.028 (0.012) for the rural share, and an imprecise 0.015 (0.010) for distance to Buenos Aires. Under instrumental variables the signs are unchanged but the estimates attenuate, and the first stage for the market-access level weakens to an F near 7 once the interaction enters, because the instrument set must now separate two endogenous regressors. We therefore read these results as sign patterns consistent with the market-access channel mattering most in agricultural, less urbanized, and remote districts, and not as magnitudes. A design with stronger identification of the interactions is left for future work.

8 Discussion

8.1 Fiscal Consolidation as Regional Policy

The reform studied here was not designed as regional policy. Section 2 documents that the closures followed fiscal criteria—operating deficits and traffic density—while the road program followed its own engineering logic. Yet the combined shock reallocated market access on a national scale: 90.7 percent of districts gained market access between 1960 and 1986, because the road expansion's gains outweighed the rail contraction's losses nearly everywhere (Figures 2 and 3). A policy debate framed as "closing unprofitable lines" was, in its spatial incidence, a large redistribution of access to markets, with winners and losers that the fiscal criteria never examined. This is the sense

[10]Estimates in this subsection are produced by the replication archive's heterogeneity program and are reported in its output files; a tabulated version is deferred until the section structure is final.

in which fiscal consolidation acted as regional policy without being one, and it suggests that the spatial incidence of infrastructure retrenchment deserves the same scrutiny routinely applied to infrastructure expansion.

8.2 Market-Access Elasticities in Comparative Perspective

The closest like-for-like benchmark for our population estimate is Gibbons et al. (2024), who study the British rail disinvestment of the same era and estimate a population elasticity of approximately 0.3. Our full-sample point estimate of 0.052 (0.031) is below that benchmark and only marginally distinguishable from zero ($p = 0.096$). The market-access approach itself originates with Donaldson and Hornbeck (2016), whose headline object—agricultural land values—capitalizes access rather than measuring a population response, so their estimates inform our method rather than our magnitudes; output-side responses to access, as in Ahlfeldt and Feddersen (2018) for German high-speed rail, are closer in kind to our sectoral estimates than to the population response.

Three measurement explanations for the low population estimate were examined and set aside. Re-anchoring market access at interior points or at each district's largest settlement does not raise the estimate, and spatial standard errors leave the headline inference unchanged (both reproduced by the replication archive's reference-point, urban-anchor, and spatial-standard-error programs; output files accompany the package). The transshipment bound, reported in Section 5.5, reshapes the market-access distribution without changing the population coefficient's order of magnitude. A candidate substantive explanation, which the research design cannot test directly, is the bimodal margin itself: the simultaneous road expansion may have cushioned the population response to rail-driven access losses. We state it as a hypothesis.

What does move the level is the trade elasticity θ: as Table 21 in Section 5.5 shows, the population coefficient ranges from roughly 0.36 at $\theta = 1$ to 0.01 at $\theta = 12$, and our main value $\theta = 4.55$ sits in the interior of that range. The comparison with the literature is therefore partly a comparison of calibrations, and we treat the level of the population elasticity as provisional on that choice. One finding is robust to it: the sectoral contrast documented below holds at every value of θ we examined (Table 22).

A second difference from the benchmarks is not about calibration: the spatial granularity of the outcome differs sharply. Our unit of observation is the district: 312 units averaging 8,140 km^2, each typically containing several separate localities — the 1960 census records approximately 3,000 localities nationwide (Dirección Nacional de Estadística y Censos, 1960b), many aligned along the rail network. Population reallocation between localities within a district, including the decline of small rail-side towns, nets out of a district-level outcome by construction. Gibbons et al. (2024) estimate their elasticity on much finer spatial units, where such reallocation is part of the measured response. A settled-population argument — mobility costs and accumulated local capital damp responses in mature economies — applies to the British episode as much as to ours and therefore cannot explain the gap between the two estimates; the aggregation margin is consistent with it. Testing it requires sub-district data, which we leave to future work.

The single-mode character of the benchmarks matters as well. Gibbons et al. (2024) study rail alone; our treatment is the joint change from a rail contraction and a road expansion measured in one multimodal index. Structural estimates for the contemporary United States indicate that the multimodal margin is quantitatively important: Fuchs and Wong (2026) find that ignoring multimodal flexibility understates the welfare gains from highway improvements by roughly a fifth in their counterfactuals. In our setting the analogous statement is descriptive rather than structural: the road expansion generated market-access gains that more than offset the rail losses for most districts (Section 3), so a rail-only account of the episode would misstate both the sign and the geography of the access change.

8.3 Sector-Mode Specialization

The clearest elasticities in this paper are sectoral, not aggregate: manufacturing production value responds at 0.317 (0.120) and manufacturing wage mass at 0.378 (0.124), while agricultural outcomes show no detectable response (Table 12). The cost schedule of Baumgartner and Palazzo (1969) offers a candidate explanation, stated in Section 2.4 as a hypothesis: rail is the cheaper mode above roughly 500 to 1,000 tons per day of cargo density and road below it (Figure A1). A modal shift from rail to road therefore lowered transport costs for the low-density, higher-value shipments characteristic of manufacturing, while raising them for the bulk flows characteristic of agriculture. Three exhibits line up behind this reading without any one of them proving it. The sectoral contrast survives measuring each sector's market access under its own cost schedule and strengthens for manufacturing (Table 11); the aggregate population elasticity rises monotonically as the cost schedule shifts from rail-favouring to road-favouring (Table 10); and the counterfactual decomposition of Section 6 now includes the sectoral outcomes (Table 25): manufacturing responds through the rail-loss and the road-gain channels alike, with estimates that are not statistically distinguishable across channels, while agriculture responds through neither. What the alignment does not establish is the mechanism's exclusivity: the matched-schedule measures strengthen identification as well as fit (their first stages are stronger), so part of the sharpening may reflect measurement rather than economics — though agriculture's matched schedule also strengthens its first stage and the agricultural response stays null, so stronger identification alone does not manufacture a response. Distinguishing the mechanisms fully would require freight flows by sector and mode, which do not exist at the district level for this period.

8.4 Limitations

Four limitations qualify the results. First, the pre-trends placebo is not a clean null: post-reform market-access changes predict 1947–1960 population growth at the five-percent level in the OLS and combined-IV specifications on the 237-district placebo subsample (Table 4), and spatial standard errors do not soften that rejection. The placebo subsample differs from the full sample, so selection and pre-trend explanations cannot be separated with available data; readers should treat the aggregate population estimates with corresponding caution. Second, three of the four outcomes in Table 17 fail the overidentification test, so for those outcomes the two instruments do not identify a common parameter; we report what each implies separately rather than reading the combined estimate as the result, and the recent-migration case is set out in Appendix A. Third, the hypothetical-road instrument is weak for total market access once road connectors are re-costed, so identification of the road component leans on functional form more than we would like; the rail component, by contrast, is strongly identified throughout. Fourth, market access is a partial-equilibrium object: it holds destination populations fixed and abstracts from price responses, so the estimates are reduced-form gradients, not general-equilibrium counterfactuals.

9 Conclusion

Argentina closed roughly ten thousand kilometers of railway between 1960 and 1986 while its paved road network nearly tripled between 1954 and 1986, a rare episode in which a country simultaneously contracted one national transport network and expanded another. We measure the resulting change in access to markets for all 312 districts with a single multimodal index, instrument it with the Larkin Plan's technical study designations and with hypothetical road networks, and estimate the response of population and economic activity on the districts with census coverage.

Three findings summarize the evidence. Aggregate population reallocation was modest: the full-sample elasticity is small, significant at the ten-percent but not the five-percent level, and sensitive to the trade-elasticity calibration. Economic specialization was not: manufacturing production and wages responded strongly where market access rose, while agriculture did not respond, a contrast robust to every calibration we examined and consistent with the modal cost structure that made roads cheap for low-density cargo. And the channel decompositions attribute the population response in similar parts to rail and road shocks—with the rail component far better identified—and roughly half of it to infrastructure located in the district itself.

For policy, the episode illustrates that decisions taken on fiscal grounds can carry first-order spatial consequences: the geography of market access was redrawn without any actor being charged with evaluating that redistribution. For research, the results caution against transferring market-access elasticities across contexts without attention to the trade-elasticity calibration, to the spatial granularity at which outcomes are measured, and to whether the underlying shock expands access into growing economies or reallocates it within settled ones.

Four extensions follow directly. A population-weighted urban anchor built from geocoded census records would settle the reference-point question definitively. On the sector-mode question, the cost side is now measured — each sector's market access under its own cost schedule (Table 11) — and the remaining steps are the demand side, weighting destinations by sectoral economic mass rather than population, and ultimately freight flows by sector and mode. Richer within-district data—stations and depots rather than kilometers—would sharpen the local decomposition that Section 7 begins. And the digitized multimodal networks and mode-specific cost schedules assembled here provide the inputs for calibrating a structural model of multimodal routing with terminal costs (Fuchs and

Wong, 2026) to the episode, which would turn the accounting decompositions of Section 6 into equilibrium counterfactuals.

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A Recent Migration

Section 5.4 reports that the two-instrument specification for the recent-migration share fails the overidentification test (Sargan $p = 0.006$; identification-robust $p = 0.012$). This appendix records the estimates and the readings that have been proposed for the sign, so that the pattern is on the record without being treated as a result.

The OLS estimate is -0.008 (0.002) and the combined IV estimate is -0.022 (0.008), both negative and significant at the one-percent level. Taken separately, the Larkin Plan instrument gives -0.007 (0.010, $p = 0.518$), with an Anderson–Rubin confidence set of $[-0.0303, 0.0138]$ that contains zero, while the hypothetical-network instrument gives -0.064 ($p = 0.029$). The instrument that Section 4.4.3 shows to be the stronger of the two therefore implies no relationship, and the negative combined estimate lies between two estimates that the data reject as descriptions of a common parameter.

Were the negative sign taken at face value, it would run opposite to a naive prediction that market-access gains attract in-migrants, and two readings would be consistent with it. First, high-market-access districts may have retained their existing populations, producing lower recent-migrant shares because turnover was low. Second, districts that lost population through out-migration to high-market-access receiving districts may have ended with small remaining populations among whom a given absolute inflow of recent migrants generates a higher share. Distinguishing these readings would require data on migration flows by origin and destination, which the cross-sectional sample does not contain. We record both possibilities and draw no conclusion.

Appendix Tables

Table A1: Descriptive Statistics, Estimation Sample

	Mean	SD	Min	Max	N
Outcomes: population (1960–1991)					
$\Delta \ln$ total population	1.025	0.539	0.051	3.104	311
$\Delta \ln$ urban population	0.868	0.515	-0.072	2.987	286
$\Delta \ln$ rural population	1.329	0.911	-1.118	6.332	272
Δ urban share	-0.045	0.193	-0.706	0.832	311
Outcomes: sectoral censuses					
$\Delta \ln$ mfg. establishments	-0.216	0.675	-2.431	4.243	310
$\Delta \ln$ mfg. production value	3.887	1.372	-1.971	9.746	310
$\Delta \ln$ mfg. wage mass	3.529	1.380	-1.703	7.208	309
$\Delta \ln$ farms	-0.254	0.497	-3.353	2.293	297
$\Delta \ln$ farmed area	0.077	0.656	-3.222	3.764	297
Outcomes: IPUMS (1970–1991)					
Δ college share	0.008	0.008	-0.008	0.049	311
Δ secondary share	0.077	0.034	0.006	0.239	311
Δ recent-migrant share	0.117	0.099	-0.098	0.583	311
Δ employment rate	0.061	0.063	-0.355	0.289	311
Placebo					
$\Delta \ln$ population 1947–1960	-0.087	0.573	-2.367	1.690	237
Treatment and instruments					
$\Delta \ln \text{MA}^{\text{full}}$	1.512	2.454	-7.592	9.531	311
$\Delta \ln \text{MA}$, Larkin instrument	-1.218	2.231	-12.578	-0.000	311
$\Delta \ln \text{MA}$, hypothetical-road instrument	-0.467	2.440	-15.732	8.847	311
Controls					
Baseline $\ln \text{MA}$ (1960)	-48.804	6.018	-62.883	-28.915	311
Baseline ln population (1960)	9.947	1.120	6.974	13.405	311
Elevation (std.)	0.002	1.001	-0.602	5.201	311
Ruggedness (std.)	0.002	1.001	-0.587	5.697	311
Wheat suitability (std.)	0.005	0.998	-1.671	2.830	311
Pre-1500 caloric potential (std.)	-0.002	1.001	-1.892	1.010	311
Post-1500 caloric potential (std.)	-0.003	1.001	-2.016	1.052	311
Distance to Buenos Aires (std.)	0.005	0.998	-1.521	3.788	311

Notes: Estimation sample (311 districts; Capital Federal excluded as an observation, Section 3). N counts non-missing observations per variable; coverage differences reflect the sectoral censuses, the 1947 placebo subsample described in Section 3, and log changes in urban and rural population being undefined for districts with zero urban or rural population in either census year. The Larkin-instrument change is nonpositive by construction (removing studied rail segments cannot raise market access). Standardised controls have mean zero and unit variance by construction over the 312 mainland districts.

Appendix Figures

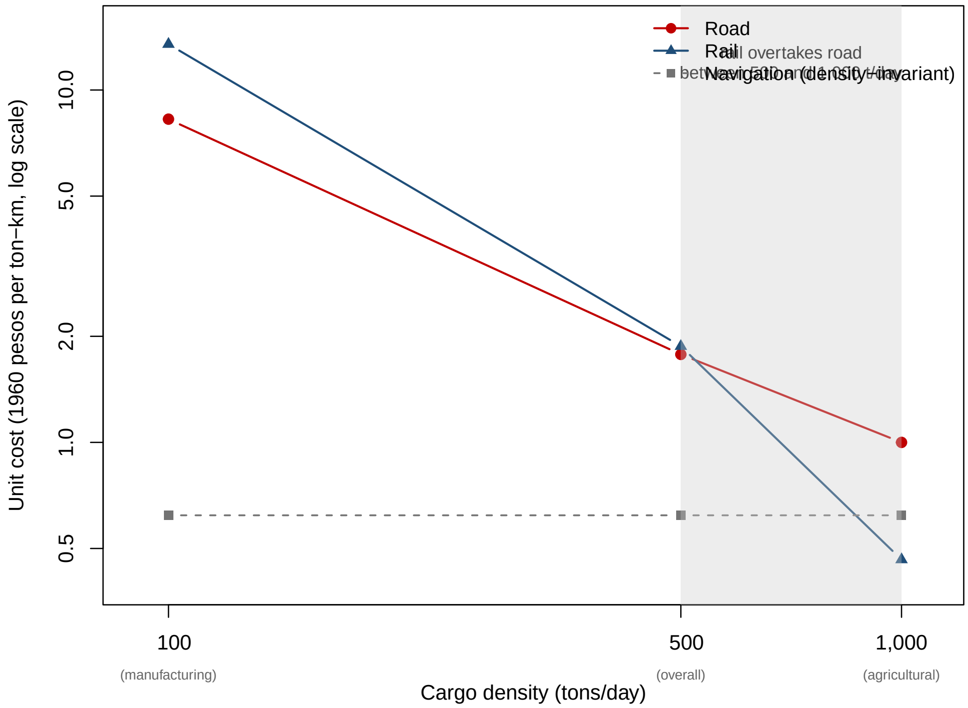

Figure A1: Transport cost per ton-kilometer by mode and cargo density, from Baumgartner and Palazzo (1969).

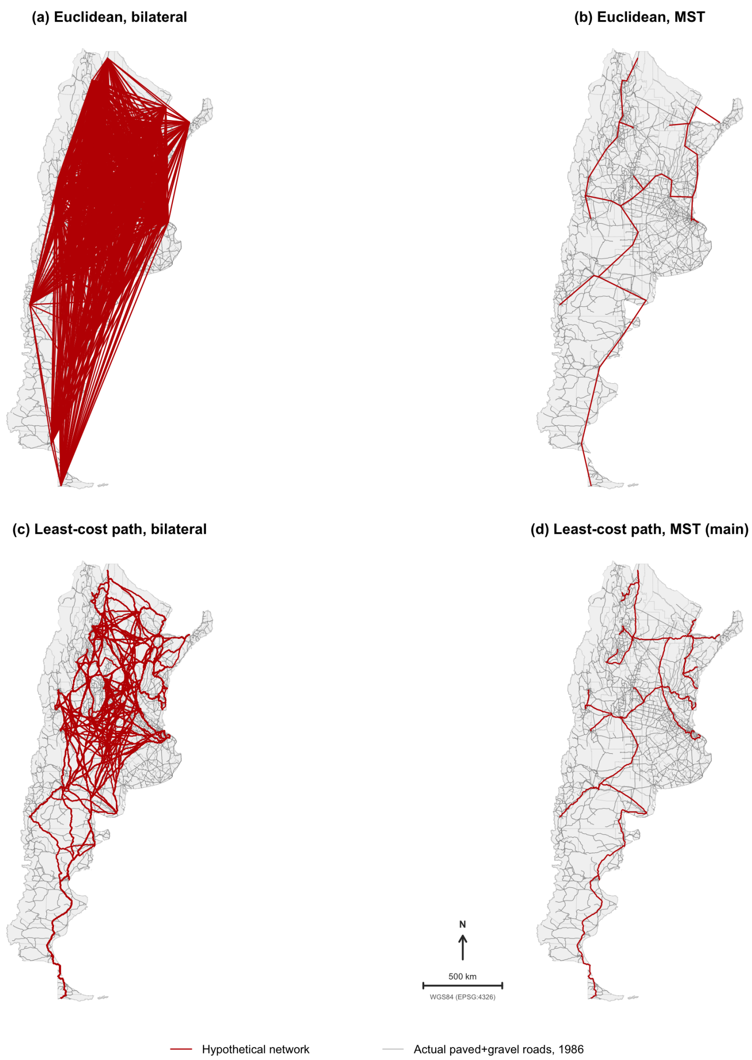

Figure A2: Hypothetical road networks used by instrument 2: least-cost-path and Euclidean variants connecting the major cities.

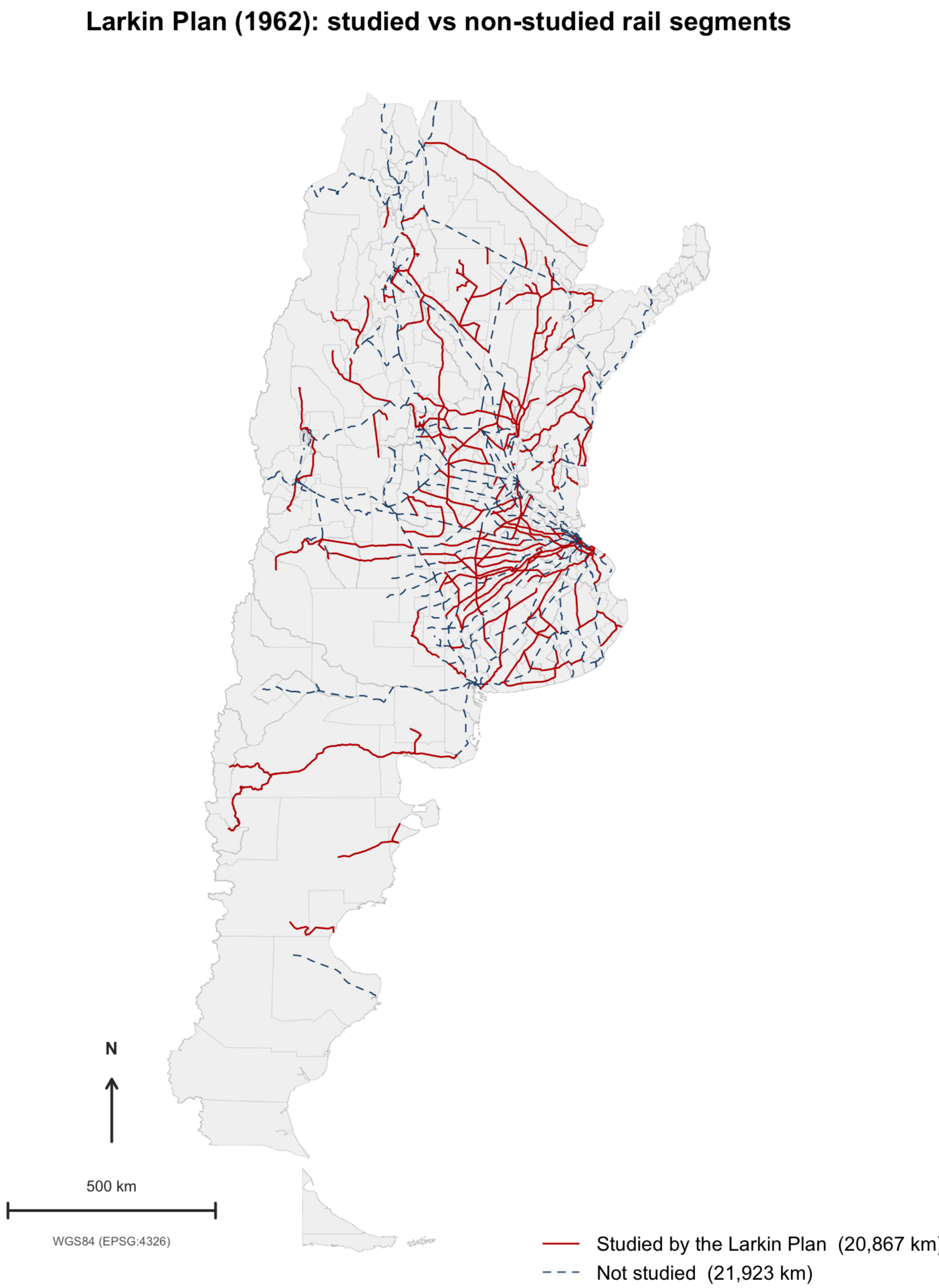

Figure A3: Rail segments studied versus not studied by the Larkin Plan, from the digitized plan tables.

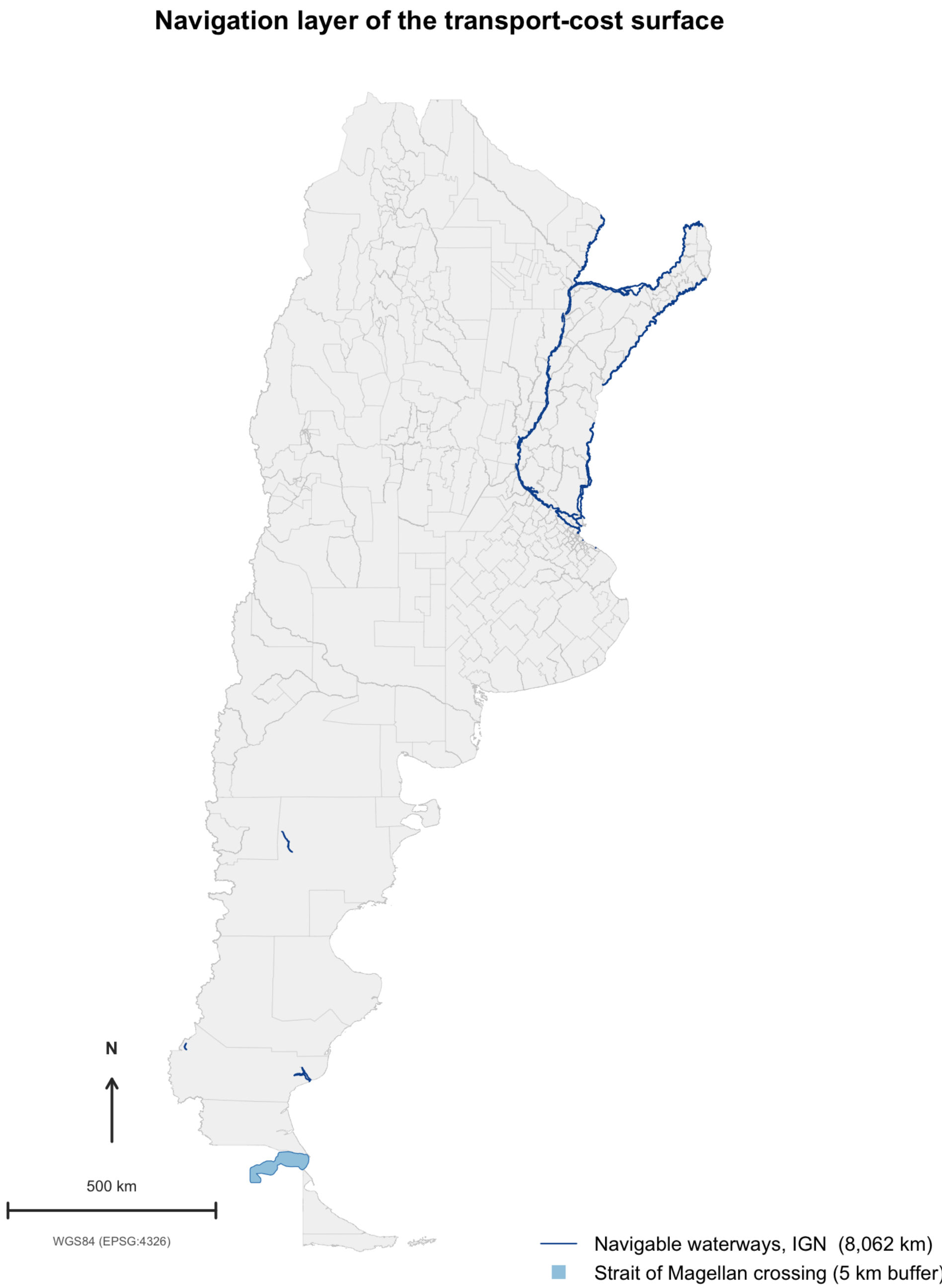

Figure A4: Navigation layer of the transport-cost surface: the navigable inland waterways flagged in the IGN hydrography (Instituto Geográfico Nacional, 2026a) — the Paraná–Paraguay–Uruguay–Río de la Plata system and a few Patagonian rivers — and the buffered Strait of Magellan crossing. Waterways are drawn as line geometries; the cost surface buffers them by 1 km when rasterising. The cost surface contains no open-ocean coastal shipping; Atlantic ports connect through land or river legs only.